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**From Statistics to Machine Learning:
A Comparative Study of Outlier Detection Methods
in Multidimensional Data**

Master of Science in Business Administration

Chair of Statistics and Data Science

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This thesis undertakes a comparative performance analysis of three outlier detection methods: the Mahalanobis distance using FAST-MCD algorithm (Rousseeuw and Driessen, 1999), the k-Nearest Neighbors (kNN) approach (Ramaswamy et al., 2000), and the contrastive learning–based method proposed by Shenkar and Wolf (2022). Outlier detection constitutes an essential component across scientific research and industrial applications. The presence of anomalous data points can compromise statistical inference. Specifically, outliers possess the capacity to inflate estimates of variability (e.g. variance and standard deviation), consequently widening confidence intervals. This effect increases the probability of committing a Type II error (failure to reject a false null hypothesis). Conversely, in certain domains, the outliers themselves represent the primary interest. Prominent examples include the application in financial fraud detection (Ramaswamy et al., 2000) and in medical diagnostics for identifying pathological conditions, such as cervical cancer (Ijaz et al., 2020).

The primary objective of this investigation is to systematically evaluate the relative efficacy of the three selected algorithms in identifying outliers within a designated dataset. Following the formal definition and characterization of outliers, the study will detail the underlying mechanisms of the three analyzed methods. This paper will begin with the robust statistical method of FAST-MCD. Subsequently, the analysis will proceed to discuss the distance-based method of kNN. Finally, the Intercont approach, a novel machine learning method will be presented. Different performance metrics will be discussed to supply a optimal view of the detection performance. The empirical investigation will utilize financial datasets, drawing among other things primary research literature. The study is guided by the following formal hypotheses regarding detection efficacy: The kNN algorithm is hypothesized to yield the highest anomaly detection performance among the three candidates. The contrastive learning–based Intercont algorithm is anticipated to outperform the FAST-MCD method but is expected to demonstrate lower efficacy than the kNN algorithm.

1 Introduction

Outlier analysis is a critical task in both academic research and industrial applications. In recent years, representation learning has been increasingly adopted to automate and refine outlier detection. A notable contribution to this field is the InterCont method, an approach based on internal contrastive learning proposed by Shenkar and Wolf (2022). While outlier detection is fundamentally a classification task, it is characterized by extreme class imbalance (Saito and Rehmsmeier, 2015, p. 2). Outliers are observations that deviate so significantly from other observations as to arouse suspicions that they were generated by a different mechanism (Hawkins, 1980, postfix). Depending on the research context, outliers may either represent operative failures (recording errors, misreporting, sampling errors) or primary targets of interest that provide critical information (Smiti, 2020).

For example in cybersecurity, it facilitates the detection of network intrusions or malware through unusual traffic patterns(Ahmed et al., 2016, Chapter 3). In finance, these methods identify fraudulent activities, such as the unauthorized use of credit cards (Malini and Pushpa, 2017). In healthcare, outlier detection assists in diagnosing rare diseases and identifying anomalous patient data (Wang et al., 2019, Chapter 2.3.).

Taxonomically, outliers are categorized into three primary types (Smiti, 2020, Chapter 2.2.1.):

- Global Outliers: Individual data points that deviate significantly from the entire dataset.
- Contextual Outliers: Data points considered anomalous only within a specific context.
- Collective Outliers: A group of data points that deviate from the expected norms of the dataset.

Distinct from outlier detection are related tasks such as Noisy Data Identification, which addresses incorrect or missing values. Novelty Detection, which seeks previously unobserved but potentially valid patterns Domingues (2019). Out-of-Distribution Detection (OOD), which identifies data points that do not belong to any of the known training classes (Yang et al., 2024).

High-dimensional datasets introduce the Curse of Dimensionality (Scott, 2015, p. 220), which is a significant challenge to the detection tasks. As dimensionality (p) increases, the volume of the space grows exponentially, causing data points (n) to become sparsely distributed. This sparsity increases the mean distance between points, making the Euclidean distance unreliable. Furthermore, in cases where $p > n$, perfect multicollinearity may occur, where variables become linear combinations of others, leading to redundancy. High dimensionality also increases the risk of overfitting, where models capture stochastic noise rather than a generalizable signal (Altman and Krzywinski, 2018).

The exceptional performance of the k-nearest neighbors (k-NN) algorithm in the analysis by Shenkar and Wolf (2022), where it surpassed the more complex architecture of InterCont in terms of F1 Score, serves as the primary catalyst for this research. While F1-Score provided the initial insight, this thesis employs ROC AUC and AUPRC as the primary metrics to ensure a more transparent and threshold-independent valuation of the methods (see Section 4.2).

The fact that a distance-based approach can yield superior results compared to sophisticated representation learning methods raises critical questions regarding the necessity of algorithmic complexity in outlier detection. This discrepancy between theoretical sophistication and practical efficacy directly inspires the following hypotheses:

- H1: k-NN will achieve the highest predictive performance across AUPRC.
- H2: While InterCont will outperform MCD, its AUPRC will remain secondary to k-NN.

While Shenkar and Wolf (2022) demonstrated k-NN's superiority via the F1-Score at a specific operating point, this thesis investigates whether this advantage persists globally across the AUPRC and ROC AUC. By shifting the focus to ROC AUC and AUPRC, it will be tested if k-NN's distance-based logic inherently provides a more robust ranking of outliers than the representations learned by InterCont.

To further evaluate the necessity of modern machine learning architectures, this research incorporates the Minimum Covariance Determinant (MCD) as a traditional statistical benchmark. MCD serves as a lower bound of algorithmic complexity, representing a robust approach to multivariate outlier detection that accounts for the masking effect (Hardin and Rocke, 2005), where outliers distort classical estimators. By establishing this baseline, the study can discern whether the performance gains of k-NN or InterCont are substantial enough to justify their increased computational overhead. Furthermore, MCD provides a global distributional perspective, offering a vital contrast to the local neighborhood analysis of k-NN and the high-dimensional feature extraction of InterCont.

This thesis will proceed with the following: In chapter 2 the theoretical foundations of machine learning, beginning with a brief description of neural network architecture before transitioning to the advanced topic of contrastive learning. Thereafter the InterCont method will be explained. The focus remains on the architecture of this method and the mechanisms through which it identifies outliers. The subsequent chapter 3 introduces the first benchmark method, k-nearest neighbors (k-NN). This section explains the underlying mechanics and architecture of k-NN while justifying its selection as a benchmark. In the following chapter, the Minimum Covariance Determinant (MCD) is introduced. It details how this method detects outliers and provides the rationale for its use as a benchmark against Internal Contrastive Learning.

In chapter 4, the dataset is initially examined by describing its primary characteristics, which serve as the basis for interpreting and explaining the results. The analysis begins with a univariate approach, utilizing boxplots to describe parameters such as the mean and distribution. Subsequently, the analysis proceeds to a bivariate approach, where the data is described using a correlation heatmap. The final stage of the analysis involves characterizing the geometry of the dataset through UMAP. The most significant metrics used in outlier detection research, facilitated by the confusion matrix, are described alongside a discussion of their respective advantages and disadvantages. The selection of metrics is finalized with the F1-Score, AUC ROC, and AUPRC. The section 5 presents the findings, followed by the section 6, where the results are explained and interpreted and limitations are illustrated. The thesis concludes with the main findings of this thesis.

2 Representation Learning for Outlier Detection

Shenkar and Wolf propose a contemporary framework for outlier detection that integrates representation learning with contrastive objectives. Within an autoencoder architecture, the method transforms input data to extract influential features for outlier detection. To establish the theoretical foundation for the experimental design detailed in Chapter 4.3, the primary paradigms of machine learning are first presented. Subsequently, the conceptual basis for InterCont is established through a concise explanation of the core components of contrastive learning.

Machine learning paradigms serve as overarching frameworks for interpreting data (Sarker, 2021, p. 4), providing the foundational logic for how algorithms process information. While individual algorithms act as specific components, the paradigm defines the broader strategy for learning. Consequently, many methods can be adapted across different paradigms depending on the data structure and the intended objective. Traditionally, the field is categorized into three primary paradigms: supervised learning, unsupervised learning and semi-supervised learning . However, numerous specialized subtypes exist, such as reinforcement learning, multi-label, and transfer learning. To establish the necessary technical background, it is essential to categorize the traditional learning frameworks and identify the specific paradigm utilized by this method.

Supervised learning is the most widely utilized paradigm, primarily employed for classification and regression tasks. It relies on labeled datasets, where a model learns a mapping from inputs to specific targets (LeCun et al., 2015). In this task-driven approach, the model is trained on labeled data, where each input is paired with the ground truth (Sarker, 2021, p.3). The model's objective is to learn a mapping function that can accurately predict labels for new, unseen data. While this paradigm achieves high performance for defined tasks, it is constrained by the high cost of manual data labeling and a tendency to struggle with out-of-distribution data.

In contrast, unsupervised learning is a data-driven approach that utilizes unlabeled data by identifying underlying patterns in data without explicit labels, making it highly scalable but often less precise for specific classification tasks. Without predefined output labels or external supervision, the model must explore the data to discover hidden patterns, underlying structures, or natural groupings (Sarker, 2021, p.4). This paradigm is highly scalable and effective for identifying complex relationships within large volumes of data without the need for expensive human annotation. However, because there are no ground truth labels to measure against, these models can be less accurate than supervised versions, and their results are often more difficult to interpret and computationally intensive.

Semi-supervised learning, utilizes labeled data in conjunction with unlabeled data. The primary objective is to leverage the structural information inherent in the unlabeled samples to improve the model's generalization performance beyond what could be achieved using the labeled data alone. This approach is particularly valuable in scientific research where acquiring labeled instances is costly, while unlabeled data is readily available. By assuming that data points in close proximity share the same label, semi-supervised algorithms can propagate information from the labeled subset to the unlabeled majority, thereby refining the decision boundaries of the model.

The authors of InterCont characterize their method within the self-supervised learning paradigm (Shenkar and Wolf, 2022, p. 3). This approach generates its own supervisory signals from the data structure itself (Jaiswal et al., 2020). Nevertheless, the network is trained on 50% labled normal data and tested on the other 50% of data that includes all labeled outliers. For this study, the same epxerimental setup will be implemented for the benchmark methods. In the proceeding thesis, this will be named semi-supervised.

2.1 Neural Networks

Artificial neural networks (NN) are computational models, designed to handle tasks such as classification, pattern recognition, and predictive modeling. A core objective in developing these networks is to achieve high generalization (Zhang et al., 2021, p. 1), ensuring that the model performs accurately on unseen test data.

The architecture of a neural network is composed of layers. The input layer receives raw data, where each node corresponds to a specific feature of the dataset. The data then progresses through one or more hidden layers. Networks with a single hidden layer are considered shallow, while those with multiple layers are termed deep networks. Deep learning (LeCun et al., 2015, p. 2) allows for the extraction of complex patterns from high-dimensional data. Within these layers, connections between nodes are governed by weights, which signify the relative importance of a connection.

During training, the network's output is evaluated against the ground truth using a loss function (Terven et al., 2025). Common loss function include Mean Squared Error (MSE), Mean Absolute Error (MAE) or Mean Absolute Percentage Error (MAPE) (Hassanat et al., 2024, p. 6). To minimize this loss, the model employs backpropagation (Rumelhart et al., 1986, p. 1). By applying the chain rule, the model calculates the gradient, hence the partial derivative of the loss function with respect to each weight.

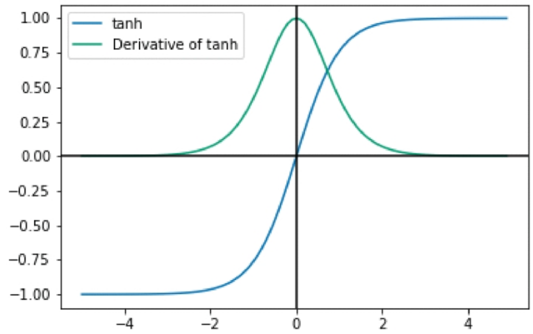
An optimizer then uses these gradients to update the parameters in the direction that reduces error. A single pass through the entire training dataset is referred to as an epoch.

Activation functions (Sharma et al., 2017, p. 311) are essential for introducing non-linearity into the model, allowing it to approximate complex functions. For a network to learn through backpropagation, these functions must be differentiable. The architecture of the InterCont method uses Tanh and LeakyReLU activation functions. They are defined as follows:

Hyperbolic Tangent (tanh), defined as

$$z = \frac{e^x - e^{-x}}{e^x + e^{-x}}. \tag{1}$$

This function is zero-centered with an output range of -1 to 1. It is susceptible to vanishing gradients.



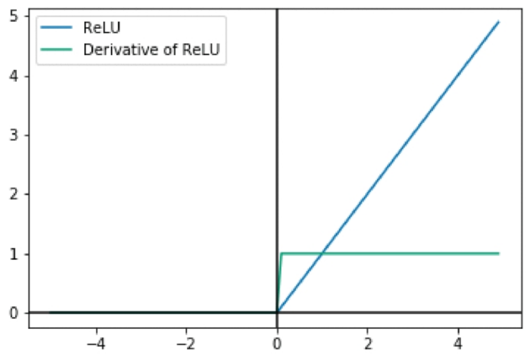
Source: (Rasamoelina et al., 2020, Fig. 4)

Figure 1: Hyperbolic Tangent Activation Function.

Rectified Linear Unit (ReLU), defined as:

$$z = \max(0, x).$$

ReLU addresses the vanishing gradient problem by maintaining a constant gradient of 1 for all positive inputs. To prevent dying ReLU, where nodes become permanently inactive due to negative inputs. Variants like LeakyReLU are used, which allow a small, non-zero gradient for negative values.



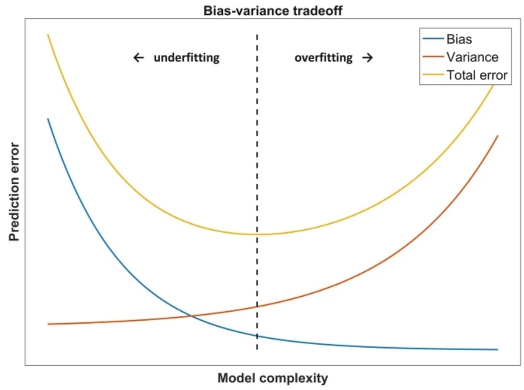
Source: (Rasamoelina et al., 2020, Fig. 6)

Figure 2: ReLU Activation Function.

2.2 Contrastive Learning

Contrastive learning represents a paradigm that has gained significant prominence within the fields of computer vision and natural language processing (Hu et al., 2024). The primary objective of such a network is to differentiate between individual instances by learning discriminative features without the necessity of manual labeling. This process

A primary challenge in training is the bias-variance tradeoff (Geman et al., 1992, p. 14). Underfitting (high bias) occurs when a model is too simple to capture the underlying data patterns. Overfitting (high variance) occurs when a model is overly complex and learns the stochastic noise of the training data rather than the generalizable signal. Techniques to mitigate these issues include increasing data volume, implementing dropout layers, or using early stopping to maintain the optimal balance between bias and variance.



Source: (Dankers et al., 2018, Fig. 8.3)

Figure 3: Bias-Variance Tradeoff.

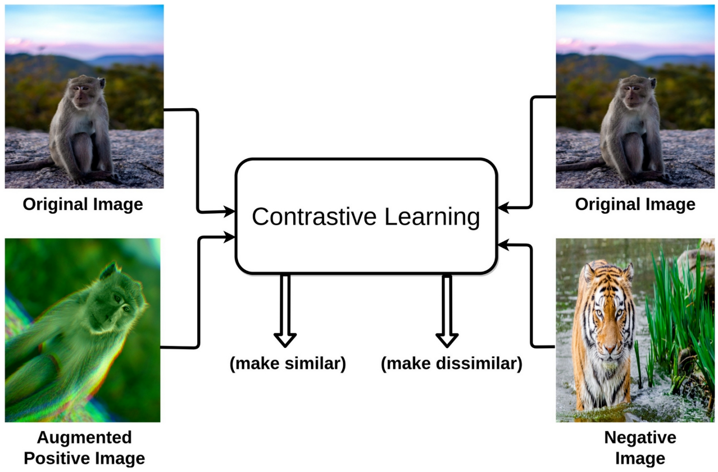
is fundamentally supported by data augmentation, which allows the model to generate multiple perspectives of the same instance. The framework introduced by Shenkar and Wolf adapts these principles for tabular data through an internal contrastive learning algorithm. The architectural design of this method facilitates the extraction of complex relationships while distilling the data into its most salient components.

Central to this approach is a contrastive loss function, which is mathematically formulated to minimize the distance between an anchor and a positive sample in the feature space while simultaneously maximizing the distance from dissimilar negative samples. This section initially delineates the data augmentation techniques prevalent in computer vision to illustrate the conceptual foundation. Subsequently, the discussion transitions to the specific architectural components of the InterCont method to examine its functional characteristics.

Data Augmentation

Data augmentation plays a critical part for the network to gain discrimminative power (Hu et al., 2024, Chapter 3.1). For example to differentiate images from apes from images of other animals, the model needs to learn main caracteristics of an ape. The model is provided with an anchor (here ape), which represents a specific data instance, such as an image, that serves as the reference point for differentiation.

A positive sample is generated by applying random data augmentation techniques to this anchor. In the context of computer vision, common augmentation strategies include cropping, flipping, and rotating. Since the anchor and the positive sample originate from the same source, the main caracterstics stay the same, hence the objective for the model is to map them to proximal positions within the feature space.



Source: (Jaiswal et al., 2020, Fig. 1)

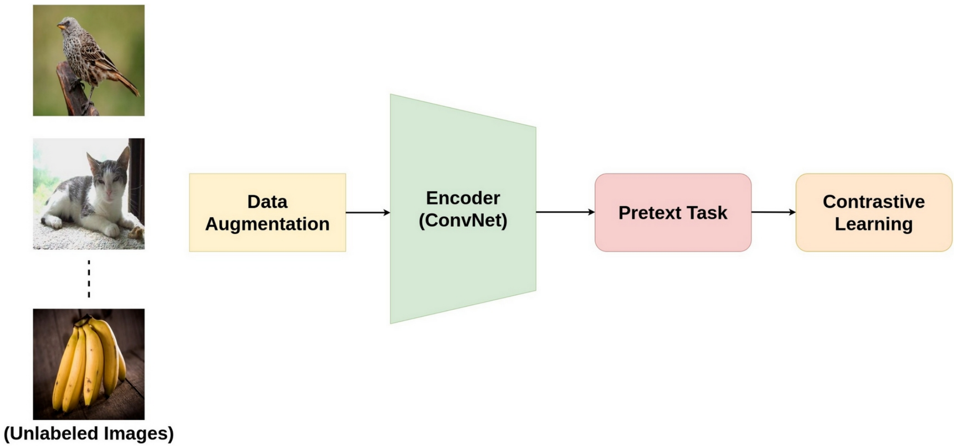
Figure 4: Intuition behind Contrastive Learning.

Conversely, a negative sample is a distinct data instance. The optimization goal is to maximize the distance between the anchor and negative samples in the feature space relative to the distance between the anchor and positive samples.

Selecting appropriate negative samples remains a significant challenge, as they must be sufficiently informative to provide the necessary signal for the network to learn diverse and robust representations. Hard negatives, which are semantically similar and lie close to the anchor in the feature space, are particularly valuable for this process. By training on a diverse set of such samples, the model enhances its ability to identify the subtle features that distinguish different objects. This mechanism mitigates the risk of overfitting and reduces the impact of data biases, thereby improving the generalization performance of the model.

Architecture

An autoencoder network comprises a neural network architecture designed to transform raw input data into a specialized feature space and latent embeddings (Mienye and Swart, 2025). This architecture typically incorporates multiple hidden layers with varying dimensionalities. The initial hidden layer projects the input vector onto a higher-dimensional space by increasing the number of nodes, which enhances the capacity of the model to represent complex relationships within the data. Subsequently, a second hidden layer compresses this representation into a vector with fewer nodes than the preceding layer. Due to this reduced dimensionality, the network is unable to retain all incoming information and is forced to discard redundant variables. This process facilitates noise reduction and ensures the preservation of the most salient patterns. Finally, an output layer aims to reconstruct the original input dimensions from the compressed representation. The network is trained by minimizing the reconstruction error, a process that ensures the encoder retains only the most critical information necessary to approximate the original signal.



Source: (Jaiswal et al., 2020, Fig. 2)

Figure 5: Contrastive Learning pipeline.

Contrastive loss function

This thesis focuses specifically on InfoNCE, as it is the loss function utilized in the methodology proposed by Shenkar and Wolf (2022). The Information Noise-Contrastive Estimation (InfoNCE) objective utilizes a density ratio to maximize the mutual information between the samples and the anchor (Oord et al., 2018). Mutual information quantifies the reduction in uncertainty regarding one random variable obtained through the observation of another (Shannon, 1948, p. 407). By framing the contrastive task as a multi-class classification problem, the model is tasked with identifying a single positive sample from a distribution containing negative samples. This mechanism encourages the encoder to extract high-dimensional features that are most salient for distinguishing related data pairs.

The loss function operates on predefined anchor and the postive sample. Specifically, the model computes a score that reflects the density ratio between the anchor and each sample, representing the likelihood that a given pair originates from the joint distribution. Consequently, the optimization process yields higher scores for positive pairs relative to negative pairs. When the model incorrectly assigns a high probability to a noise sample, the resulting loss is not minimal, necessitating weight adjustments.

Through this iterative refinement, the model ensures the positive sample achieves the highest density ratio. The tightness of the mutual information lower bound is inherently linked to the number of the negative samples. As the number of negative samples, and the incorporation of hard negatives, increases, the discrimination task becomes more demanding, leading to a more robust estimation of the underlying data structure.

2.3 Internal Contrastive Learning (InterCont)

This chapter examines the internal contrastive learning algorithm proposed by Shenkar and Wolf (2022). The primary objective is to derive an outlier score $y(x)$, which assigns low values to samples consistent with the underlying distribution and higher values to potential outliers (Shenkar and Wolf, 2022, p. 1). The following chapter details the architectural configuration of the method, beginning with the network construction and the selection of activation functions. Subsequently, the analysis addresses the specific loss function and the normalization techniques implemented to calculate cosine similarity. The operational procedure of a single epoch is then described, followed by an evaluation of the relevant hyperparameters. Finally, the architecture of this method is compared to the contrastive learning framework delineated in the preceding section.

The network architecture consists of two asymmetrical, fully connected neural networks, denoted as F and G (Shenkar and Wolf, 2022, p. 5). The authors constructed an autoencoder network as described in the chapter before. This structural asymmetry is required to accommodate the differing dimensions of the input vectors, where b serves as the anchor and a represents the positive sample. The authors determine the dimensions of the hidden layers relative to the embedding size u. Network F, designed to process the larger anchor vector b, utilizes two hidden layers containing u and $2u$ nodes, respectively. Network G maintains a similar structure with two hidden layers. However, because it processes the smaller complementary positive sample a, the node counts are reduced. Specifically, the first hidden layer of G comprises $\frac{u}{4}$ nodes, while the second hidden layer comprises $\frac{u}{2}$ nodes.

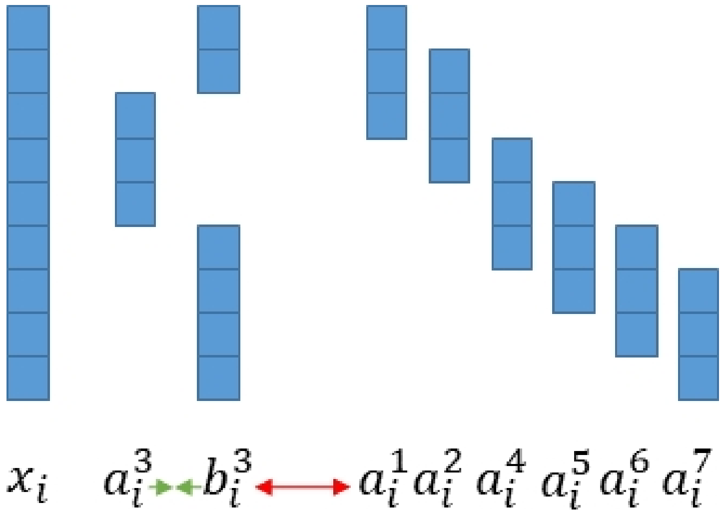
The architecture utilizes the LeakyReLU activation function with a slop coefficient of 0.2 across its layers, with the sole exception of the first hidden layer in Network F, which employs a Tanh activation. The inclusion of Tanh in the initial layer of the anchor network serves to center the input data and bound activations within the range of -1 to 1, facilitating stability during the early stages of feature extraction for the larger input vector. Furthermore, without this differentiation, the network might produce class-independent results lacking discriminative power. Consequently, a and b are mapped into distinct mathematical spaces.

The authors utilize the previously discussed InfoNCE loss function. This loss is taken as the outlier score, because samples that deviate from the underlying distribution receive higher scores, indicating an increased probability of being categorized as an outlier.

Epoch

The authors proposed a sliding window approach for data augmentation to facilitate internal contrastive learning (Shenkar and Wolf, 2022, p. 3).

The model initialization involves treating the rows of the dataset S as a vector x_i with a total length of d. The number of pairs $\phi(x_i) = (a_i, b_i)$ for one instance x_i is determined by the formula $m = d + 1 - k$, where k represents the defined length of the positive sample a_i. The anchor b_i is composed of the residual features $d - k$, functioning as the complementary vector to the positive sample.



Source: (Shenkar and Wolf, 2022, Fig. 1)

Figure 6: Sliding Window Approach.

A distinguishing feature of this method is that the negative samples are derived from the positive samples of other constructed pairs within the same data instance, which characterizes the internal nature of the contrastive learning process. For a visual representation, see Figure 6. After this iterative procedure the resulting vectores of b and a are combined into network F and G.

The networks, denoted as F^N and G^N, incorporate a double $L2$ normalization process. The initial row-wise normalization standardizes the scale of each sample, ensuring that outliers are not identified solely by the magnitude of specific features and projecting each sample onto a unit hypersphere. Subsequently, a column-wise normalization is applied to standardize the distribution of individual features across the batch.

Through the specified embedding sizes, the vectors are projected into a higher-dimensional space and reconstructed thereafter. The network aims to learn the most relevent relationships and discards noise within the data.

The the Info-NCE loss function is implemented as follows:

$$\ell(F, G, \Phi(x_i), j) = -\ln \frac{\exp(F^N(b_i^j) \cdot G^N(a_i^j)/\tau)}{\sum_{j'=1}^{m} \exp(F^N(b_i^j) \cdot G^N(a_i^{j'})/\tau)}. \tag{2}$$

The numerator of the equation quantifies the cosine similarity between the $F^N(b)$ and $G^N(a)$ vectors by calculating their dot product. A higher product indicates a greater degree of similarity between the projected vectors.

The denominator quantifies the similarity between the anchor b_i and all other positive samples a_j, which serve as negative pairs. By summing the exponentiated dot products of the anchor with these negative samples, the denominator remains positive. To minimize this loss, the negative natural logarithm is applied, ensuring the result is small when the similarity term is large. Conversely, as the similarity between the anchor and the correct positive sample decreases relative to the negative samples, the loss increases.

If $F^N(b_i) \cdot G^N(a_i) = 1$, the two vectors point in an identical direction within the feature space. A lower value signifies a lack of similarity, reaching 0 when the vectors are orthogonally placed. Conversely, the vectors are diametrically opposed reaching a minimum of -1 signifiying dissimilarity. Because $G^N(a_i)$ is distinct from $G^N(a_j)$, the cosine similarity $F^N(b_i) \cdot G^N(a_i)$ is expected to be greater than $F^N(b_i) \cdot G^N(a_j)$ The relationships of all negative pairs are aggregated to compute the total loss, and this procedure is performed iteratively for all instances within the dataset.

Through backpropagation, the gradients are calculated and the network weights are updated iteratively. To minimize the loss, the method must increase the angle between negative samples and the anchor, thereby decreasing the sum of the dot products in the denominator while maximizing or maintaining the dot product in the numerator. This optimization forces the representations of unrelated samples apart in the embedding space. The training process terminates when the loss falls below a threshold of 10^-3, signaling that the model has reached the desired level of convergence (Shenkar and Wolf, 2022, p. 5).

The sliding window approach possesses a significant limitation, as it requires dimensions with high mutual information to be positioned in close proximity to one another. This condition is frequently unmet in tabular data, where the window size may be insufficient to encompass these related dimensions. To address this, the authors implement a permutation strategy to change the order of dimensions, facilitating a bagging effect.

Consequently, the method constructs r distinct versions of S, and the training process proceeds independently for each version of the table. This strategy ensures that the model captures a broader range of dependencies across the feature space, mitigating the risk of omitting critical relationships due to arbitrary feature ordering.

The model is characterized by two primary hyperparameters (Shenkar and Wolf, 2022, p. 3):

1. k determines the number of consecutive number of the features to form the positive sample a.

2. u determines the embedding size.

If k is assigned a high value, the model risks losing its local focus and neglecting critical outlier information. Conversely, if k is excessively small, the model may fail to capture global relationships by concentrating solely on local dependencies, which can degrade overall performance. For the dataset utilized in this research, where $d = 129$, the authors determined the optimal window size to be $k = 2$ (Shenkar and Wolf, 2022, p. 5). This parameter selection serves to balance the granularity of feature interactions with the computational efficiency of the contrastive task.

The embedding size u determines the dimensionality of the space available for the method to estimate relationships between data points. If u is insufficient, the model is forced to discard information, potentially resulting in the loss of critical indicators relevant to outlier detection. Conversely, an excessively large u may lead to overfitting, where the network captures noise rather than generalizable patterns. For their implementation, the authors fixed u=200, providing a balanced capacity for feature representation without compromising model stability.

The temperature constant τ serves to scale the dot products, thereby modulating the sharpness of the probability distribution. When τ is small, the resulting values of the similarity scores are magnified. In combination with the exponential function, even marginal differences between the numerator and the denominator are accelerated, forcing the model to be more certain in its alignment of positive pairs. The authors fixed $\tau = 0.01$ to ensure that the contrastive loss remains highly sensitive to these feature representations, encouraging the network to develop a more discriminative embedding space.

The authors conducted robustness checks to evaluate the sensitivity of the model to various parameter configurations. Regarding τ and u, the analysis revealed no significant volatility in the results, suggesting that the model maintains stability across a range of values for the temperature constant and the embedding dimensionality (Shenkar and Wolf, 2022, p. 7).

3 Benchmark Methods

To analyze the performance of the InterCont method, it is essential to establish a baseline using recognized benchmark methods. Comparing a new proposal against established algorithms allows the scientific community to identify the specific conditions under which a method excels or fails. Contemporary research evaluates a vast array of techniques. For instance, Shenkar and Wolf (2022) compared InterCont against 31 datasets of the ODDS repository (Shenkar and Wolf, 2022, p. 7) Such an extensive comparison is beyond the scope of this thesis. Instead, two representative methods have been selected: one based on a statistical approach and the other on a distance-based approach.

This chapter is organized as follows:

- k-Nearest Neighbor (k-NN): Introduced as a traditional distance-based approach for outlier detection.
- Minimum Covariance Determinant (MCD): Introduced as a traditional statistical method for robust estimation.

For each method, the frameworks are defined and their respective advantages and disadvantages are discussed.

3.1 Distance-based approaches for Outlier Detection

The k-Nearest Neighbor (k-NN) algorithm is a non-parametric method that utilizes the Euclidean distances for outlier detection. Initially detailed in a technical report for the US Air Force by Fix and Hodges (1951), its formal theoretical proof was later established by Cover and Hart (1967).

The application of k-NN for outlier detection was pioneered by Knorr and Ng (1998). They defined an observation as an outlier if a specific fraction of the dataset lies beyond a predefined distance D, thereby introducing a distance-based hyperparameter. Ramaswamy et al. (2000) subsequently refined this approach by eliminating the distance threshold in favor of an internal outlier score, calculated based on the distances to an object's neighbors. Under this framework, the n points exhibiting the highest scores are classified as outliers, where n is determined by a user-defined outlier ratio.

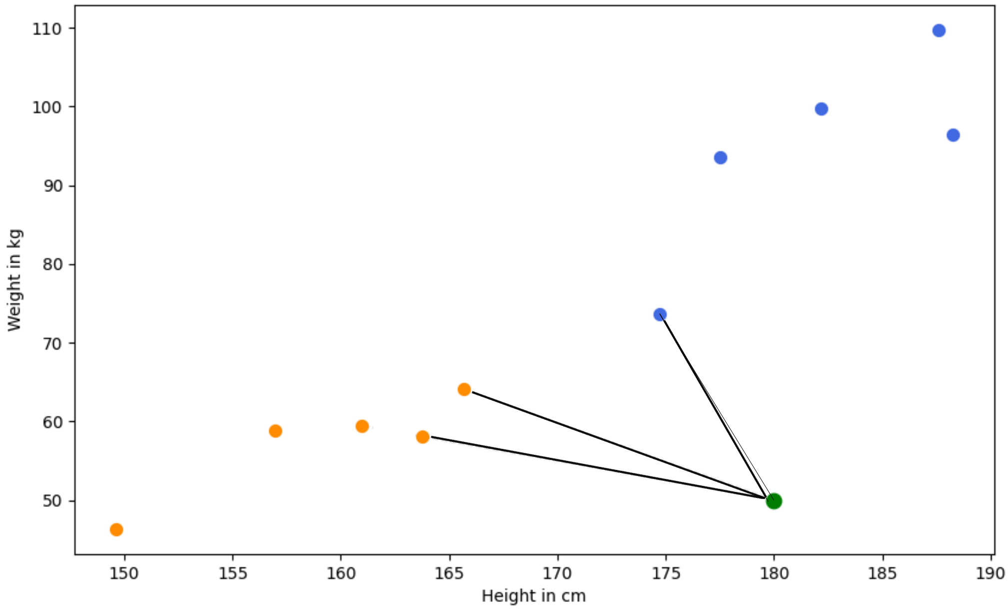


Figure 7: Grapfical Depiction of k-NN Algorithm.

While the original InterCont publication does not explicitly specify the k-NN variant employed, the authors utilize the PyOD library (Zhao et al., 2019). Given that PyOD implements the version proposed by Ramaswamy et al. (2000), this specific iteration is utilized as the benchmark for this thesis. K-NN is typically follows an unsupervised learning pradigm. Nevertheless, the experimental setup differs from this paradigm to make the benchmark comparable to InterCont.

There are different variants of k-NN, mitigating disadvantages of the original algorithm. For example by utilizing the average distance across the neighborhood enhances the robustness and stability of detected outliers (Angiulli and Pizzuti, 2002). Because this methods is constructed as a benchmark, the traditional version of k-NN wil be implemented, which is based on Euclidean distance to the specific k-th nearest neighbor. The setup is discribed in chapter 4.3.

The algorithm executes the following procedure:

1. The distances between kth datapoints are calculated.

2. The distances are ranked from highest to lowest.

3. The number of instaces n with the highest distances are defined as outliers.

(Ramaswamy et al., 2000, p. 429) k-NN is characterized as a lazy learning algorithm (Wettschereck et al., 1997, p.273), meaning it does not construct an explicit generalization or internal model during a training phase. Consequently, the entire dataset must be retained. If a new instance is introduced, the algorithm must re-evaluate the proximity metrics across the dataset to determine its status.

To determine the k-nearest neighbors, the algorithm must quantify the proximity between an instance and its neighboring data points. In traditional k-NN implementations, Euclidean distance (Abu Alfeilat et al., 2019, p. 226) is the standard metric.

It is defined as follows:

$$d(p,q) = \sqrt{\sum_{i=1}^{n}(q_i - p_i)^2}. \tag{3}$$

Where p and q represent two data points in an n-dimensional space. The squared differences between corresponding coordinates are summed, and the square root of this sum yields the Euclidean distance, which quantifies the straight-line distance between the two points in the feature space.

However, the use of Euclidean distance renders k-NN non-affine equivariant (Biau et al., 2012, p. 24). That means that the results do not remain consistent under affine transformation of the data. Those transformations include scaling, rotation, or translation. Because this metric treats all dimensions equally, variables with larger numerical ranges will dominate the distance calculation, potentially masking outliers that exist within the distribution of smaller-scaled variables. Consequently, the relative scaling of variables directly influences the identification of potential outliers.

Normalization

To mitigate the effects of variable scaling, the dataset must be normalized (Performance et al., 2021, p. 3). Min-Max Normalization is frequently employed to rescale all features to a fixed range of [0,1].

It is defined as follows:

$$x_{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}}. \tag{4}$$

Where x_i is the original value of the feature, x_{min} is the minimum value of that feature across the dataset, and x_{max} is the maximum value. This technique performs a proportional transformation that preserves the original distributional and correlational structures. However, it is sensitive to global outliers, which can compress the remaining data into a narrow, indistinguishable range.

Alternatively, Z-score normalization is utilized. This method transforms a variable such that the resulting distribution has a mean of $\mu = 0$ and a standard deviation of $\sigma = 1$.

It is defined as follows:

$$z = \frac{x_i - \mu}{\sigma}. \tag{5}$$

While effective for data approximating a Gaussian distribution, the utility of Z-score normalization diminishes if the data is heavily skewed or non-Gaussian, as the mean may not accurately represent the central tendency of the distribution . The presence of outliers could result in a distorted standard deviation σ, which complicates outlier detection due to the masking effect, where outliers inadvertently inflate the dispersion metrics used to identify them.

The Robust Scaler addresses this by substituting the mean μ and standard deviation σ with more resilient statistics: the Median and the Interquartile Range (IQR). By centering and scaling based on percentiles, the influence of outliers on the transformation process is significantly reduced.

It is defined as follows:

$$x_{robust} = \frac{x_i - \tilde{x}}{\text{IQR}}. \tag{6}$$

This method scales the data based on its internal distribution (Izonin et al., 2022, p. 4). By subtracting the median in the numerator, the data is centered such that the median corresponds to 0. Observations exceeding the median result in positive values, whereas those below it become negative.

The denominator facilitates the scaling process. By utilizing the IQR, the scaling factor is derived from a parameter that is inherently resistant to outliers. Consequently, the scaling maintains the relative distances between normal observations without the distortion typically introduced by outliers in the dataset, preserving the sensitivity of the k-NN algorithm in distinguishing outliers from the regular data structure. Therefore, the Robust Scaler will be implemented into the experimental setup, see 4.3.

Hyperparameter

In addition to the outlier threshold, the parameter k represents the primary hyperparameter of the algorithm. It defines the number of proximal observations utilized to compute the distance metrics.

An excessively low value for k typically leads to overfitting, as the model becomes sensitive to local noise. This sensitivity often results in the identification of false positives and the generation of complex, fragmented decision boundaries. While a larger k provides a smoother decision surface and increased robustness to stochastic fluctuations, a value that is too high may lead to underfitting, where the model fails to detect local outliers because they are overshadowed by the broader neighborhood. Consequently, the resulting decision boundaries become overly generalized, failing to capture critical local patterns within the data.

A statistically rigorous method to determine k involves utilizing k-fold cross-validation (Wong and Yeh, 2019, p. 1586) to optimize the trade-off between bias and variance. This ensures that k is tuned to the underlying data structure rather than selected arbitrarily. In k fold cross validation the dataset is split into k same sized subsets. The model is trained on $k - 1$ of these subsets while the remaining subset is used for testing. This process is repeated k times, with each subset serving as the test set once. The average performance across all k iterations is then used to evaluate the model's generalization. K-fold cross validation ensures that all data will be used for hyperparameter tuning.

The k-NN algorithm offers several distinct advantages (Taunk et al., 2019, p. 1258). The algorithm intuitive, facilitating straightforward interpretation of the detected outliers. While susceptible to the curse of dimensionality, the algorithm remains computationally efficient with large training sets. Its reliance on distance metrics rather than complex optimization routines ensures a user-friendly implementation. As a non-parametric model, k-NN requires no prior assumptions regarding the underlying distribution of the dataset, providing the necessary flexibility to analyze complex data structures effectively.

The computational complexity of k-NN presents significant disadvantages in large-scale applications. As a memory-intensive approach, the model must store the entire dataset to facilitate predictions, which can result in substantial latency. Furthermore, the reliance on distance-based metrics makes the algorithm susceptible to noise, which can negatively impact performance. Consequently, the selection of k is critical to the model's performance, necessitating rigorous hyperparameter tuning.

k-NN serves as a traditional machine learning algorithm. Additionally it uses a distance based approach to detect outliers and it does not need assumptions about the underlying distribution of the data. If the results of k-nn does not exceed the performance of InterCont, it could indicate that InterCont camptures the internal structure more granular.

3.2 Statistical Approaches for Outlier Detection

Statistical methodologies for outlier detection operate on a different fundamental logic than the other approaches previously discussed. Parametric methods assume the underlying distribution of the data and estimate population parameters, such as the mean and covariance Smiti (2020). This allows for statistical inference, wherein the model assigns a probability to each data point, quantifying the likelihood that it constitutes an outlier.

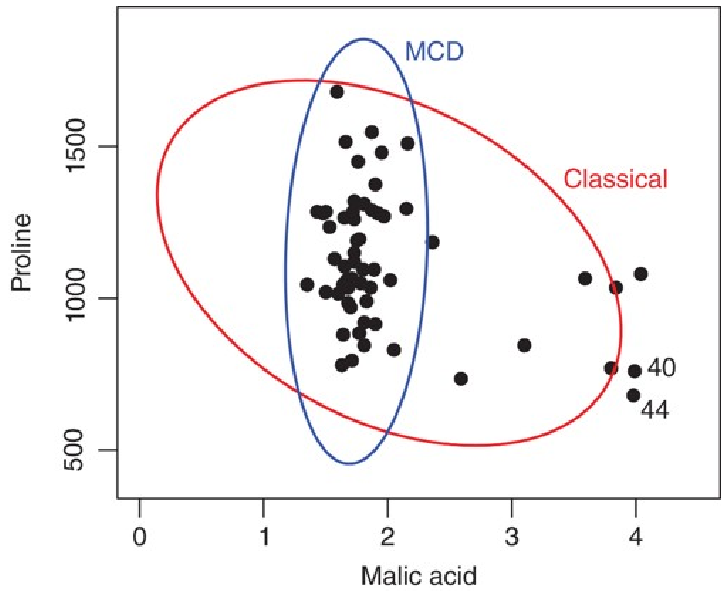
The following section outlines the structural framework of this analysis. First the framework of the Minimum Covariance Determinant (MCD) Rousseeuw (1985) estimator will be described, including an assessment of its inherent limitations. To address these computational challenges, the FAST-MCD algorithm is introduced, which serves as the primary implementation for the subsequent experiments. FAST-MCD Rousseeuw and Driessen (1999) represents a significant refinement of the original MCD approach, providing the computational efficiency necessary for practical outlier detection. While further variants such as DetMCD (Hubert et al., 2012) exist, they are excluded from this implementation to maintain a focus on a well-established and widely recognized statistical framework for outlier detection.

Following the detailed description of the algorithmic procedures, the relative advantages and disadvantages of this method are evaluated to justify its selection as a benchmark within the experimental design of this thesis.

A naive approach to outlier detection could involve estimating the sample mean and covariance matrix for the entire dataset and labeling the n observations with the largest distances as outliers. By utilizing the Mahalanobis distance (Mahalanobis, 1936), the analysis accounts for the underlying covariance structure of the data. This standardizes the multidimensional space, effectively magnifying the relative distance of outliers that might otherwise remain hidden in a standard Euclidean framework (Leys et al., 2018, Chapter 2).

However, this standard procedure is often compromised by the masking effect (Hardin and Rocke, 2005, Chapter 1). Because Mahalanobis distances are sensitive to outliers, even a single outlier can shift the mean toward itself and artificially inflate the covariance.

This effectively masks the outlier's presence, leading to false classifications. To address this, the Minimum Covariance Determinant (MCD) was proposed by Rousseeuw (1984) and theoretically formalized in Rousseeuw (1985) to provide robust estimators that remain unaffected by data contamination. It is a parametric and affine-equivariant estimator for the mean and covariance matrix of a dataset (Rousseeuw, 1985, Chapter 3).



Source: (Hubert et al., 2018, Fig. 1)

Figure 8: Minimum Covariance Determinant.

Minimum Covariance Determinant (MCD)

The objective of the MCD is to identify a subset H containing h observations that yields the minimum possible and robust covariance matrix determinant. Geometrically, the MCD seeks the smallest possible ellipsoid that encapsulates h numbers of observations. By estimating the covariance ellipsoid based on the region of highest local density, the MCD robustly identifies a representative subset of the data. This approach effectively isolates global and contextual outliers, thereby mitigating the masking effect. For a graphical depiction, see figure 8.

FAST-MCD

A significant limitation of the original MCD algorithm is its computational complexity (Rousseeuw and Driessen, 1999). Finding the optimal subset requires evaluating $\binom{n}{h}$ combinations, which is infeasible for large datasets (Rousseeuw, 1985, Chapter 4). Therefore Rousseeuw and Driessen (1999) introduced the FAST-MCD algorithm, which utilizes concentration steps (C-Steps) to iteratively improve the subset selection and pruning the process to mitigate the estimation of $\binom{n}{h}$ combinations. The FAST-MCD algorithm is as follows:

1. Determine the subset size h, using the formula $h = \frac{n+p+1}{2}$.

2. Perform C-Step on several random subsamples:

   2..1 Initialize by drawing a random subsample J with $|J| = p + 1$ observations. Calculate the sample mean and the sample covariance matrix. If $\det(S_0) = 0$, add another random observation until $\det(S_0) > 0$.

   2..2 If $\det(S_0) > 0$, compute the Mahalanobis distances for all n observations.

   2..3 Sort the Mahalanobis distances from smallest to largest.

   2..4 Select the h smallest distances to form H_2 and compute its mean and covariance matrix.

3. Select the ten results with the lowest determinants ($\det(S)$) and iterate C-Steps until convergence.

   a) Estimate μ_S and Σ_S

   b) Estimate Mahalanobis distance of all n instances.

   c) Chose h instances with the smallest Mahalanobis distance

   d) End with convergence

4. Report the mean μ^{MCD} and covariance Σ^{MCD} from the subset with the overall lowest determinant.

(Rousseeuw and Driessen, 1999, Chapter 3)

This selective iteration is efficient because the distinction between robust and non-robust solutions usually manifests within the first two steps. The primary hyperparameter is h, and it is constrained by the range:

$$\frac{n+p+1}{2} \leq h \leq n. \tag{7}$$

The lower bound is set, so the subsample contains at leats halve of the dataset $\frac{n}{2}$. By definition, an outlier is an observation that deviates from the majority of the data. If the contamination ratio exceeds 50%, the anomalous points constitute the majority class, meaning the original data distribution is no longer statistically identifiable as the normal state.

Additionally $h > p$ has to be satisfied, because the covariance matrix becomes singular $det(S) = 0$, rendering it non-invertible. To avoid excessive noise, it is recommended to maintain at least five observations per dimension (Hubert et al., 2018).

The breakdown point determines the maximum proportion of arbitrarily large outliers that a dataset can contain before the estimator produces unreliable results (Hampel, 1971, p. 1894). There is a trade-off between robustness and statistical efficiency (Rousseeuw and Driessen, 1999, Chapter 4). The higher the breakdownpoint is, the lower the statistical efficiency gets, resulting in a increased variance. If the dataset contains less then 25% of outliers, the subset can be set to $h = (0.75n)$

Once the estimators μ^{MCD} and Σ^{MCD} are calculated, the Mahalanobis distance D_i is computed for each observation:

$$D_i^2 = (\mathbf{x}_i - \hat{\mu}_{\text{MCD}})^T \hat{\Sigma}_{\text{MCD}}^{-1} (\mathbf{x}_i - \hat{\mu}_{\text{MCD}}). \tag{8}$$

Assuming the clean data follows a multivariate normal distribution, D_i^2 follows a χ^2 distribution. This allows for a formal discordancy test: if D_i^2 exceeds a critical threshold defined by a significance level α, the point is flagged as an outlier. For this thesis, a fixed threshold approach is used to ensure comparability with the other methods. The implementation of the method will be described in chapter 4.3.

In summary, the MCD is defined by its high breakdown point and affine equivariance, which ensures reliability without the need for normalization. It effectively solves the masking problem and provides transparent, quantifiable results through probability-based scoring. However, it remains computationally demanding, assumes a Gaussian distribution, and is susceptible to the Curse of Dimensionality if n is not sufficiently larger than p (Smiti, 2020, p.5).

The MCD estimator is primarily oriented toward the identification of global and contextual outliers, as it seeks a subset of observations that minimizes the determinant of the covariance matrix to define a robust location and shape of the data. Consequently, the MCD is not inherently structured to identify collective outliers. If a group of outlying observations is sufficiently large and dense, it can exert influence on the estimator, causing the resulting hyper-ellipsoid to dilate and accommodate the outlier cluster. Such a distortion of the estimated distribution could not only lead to false negatives by masking the actual outliers but also could generate false positives by excluding legitimate inliers from the robustly defined region.

MCD serves as a statistical benchmark in this study. It provides a baseline to determine whether a dataset necessitates complex machine learning models like InterCont, or if traditional robust statistics are sufficient.

4 Experiments

A major challenge in outlier detection research involves establishing comparability between the methods, which necessitates the implementation of a unified framework. Such a framework ensures that the same dataset and evaluation metrics are consistently applied. Consequently, the methods need to be implemented in such a way that a fair comparison is facilitated.

The subsequent section details the practical implementation of the conducted experiments, beginning with a comprehensive analysis of the dataset. This examination facilitates a progressive understanding of the data, transitioning from univariate and bivariate perspectives to a structural evaluation using Uniform Manifold Approximation and Projection (UMAP). Following this empirical investigation, the discourse shifts to a critical assessment of potential evaluation metrics. These metrics are argued and justified, culminating in a definitive selection based on the previously identified data characteristics. The experimental framework is defined to ensure that all benchmarks maintain rigorous comparability with the InterCont methodology. Upon establishing this foundation, the empirical results are presented.

To ensure statistical significance and validity, a Friedman test is employed during the evaluation process. Subsequently a post-hoc Wilcoxon signed-rank test is applied to rank the method' performance. These findings are subsequently interpreted within chapter 6, while the final conclusions and implications of the research are synthesized in the terminal chapter of the thesis.

4.1 Analysis of the Wine Dataset

The Outlier Detection DataSets (ODDS) library serves as a widely utilized repository for benchmarking anomaly detection algorithms (Rayana, 2016). For instance, Shenkar and Wolf (2022) employ these datasets to evaluate the performance of various outlier detection methods, including the previously discussed InterCont approach (Shenkar and Wolf, 2022, p. 5). The repository comprises 31 distinct datasets characterized by varying dimensionalities and sample sizes.

While utilizing a broad spectrum of datasets is advantageous for identifying the specific strengths and weaknesses of a method, analyzing all 31 datasets is beyond the scope of this thesis. Consequently, this research focuses on a detailed evaluation of the Wine dataset to derive insights for the results section.

The primary objective of the following analysis is to characterize the dataset's properties comprehensively. This process begins with a statistical summary encompassing central tendencies and distributional shape. This is followed by an examination of bivariate relationships between variables. Finally, the analysis addresses the geometric properties of the data, specifically focusing on its manifold structure.

The Wine dataset was submitted by Aeberhard and Forina (1992) on June 30, 1991, to the UC Irvine Machine Learning Repository. The original data consists of chemical analyses of wines derived from three distinct Italian cultivars. While initially intended for multi-class classification based on the three producers, the data was transformed into an imbalanced binary classification task for the ODDS repository. The processed version contains 129 instances and 13 attributes.

In this configuration, instances from the second and third cultivars are aggregated to form the inlier class (119 instances), while 10 instances from the first cultivar are defined as outliers. This results in an outlier ratio of 7.7%, which places the dataset in comparison to other benchmarks in the repository ($mean = 8.61\%$). Furthermore, the version provided by the ODDS repository contains no missing values and consists exclusively of continuous variables. To maintain visual clarity in the subsequent analysis, feature names are omitted from certain graphical representations. However, Figure 9 provides a corresponding reference for these variable names.

Feature	Dimension
Feature 1	Alcohol
Feature 2	Malic acid
Feature 3	Ash
Feature 4	Alcalinity of ash
Feature 5	Magnesium
Feature 6	Total phenols
Feature 7	Flavanoids
Feature 8	Nonflavanoid phenols
Feature 9	Proanthocyanins
Feature 10	Color intensity
Feature 11	Hue
Feature 12	OD280/OD315 of diluted wines
Feature 13	Proline

Figure 9: Table of Names.

Univariate Analysis

Figure 10 illustrates the distribution of each feature using boxplots (Tukey, 1977, p. 39), which display the mean, median, interquartile range (IQR), and whiskers defined by 1.5 × IQR. In these visualizations, the red dotted line represents the mean, while the grey line indicates the median.

The IQR represents the central 50% of the data points located between the Q_1 and Q_3. A wider IQR indicates greater variability within a specific feature, whereas a narrow IQR suggests that the observations are more densely clustered. The features within the dataset exhibit significantly different scales of dispersion. While Features 3, 4, 5, and 6 are characterized by relatively narrow IQR, Feature 12 displays a notably large IQR.

The whiskers extend to most extreme data points within $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$ respectively, where Q_1 and Q_3 represent the first and third quartiles of the data distribution. The relative lengths of the whiskers provide further evidence of distributional skewness. A top whisker that is longer than the bottom whisker indicates a right-skewed distribution, whereas the inverse suggests left-skewness. This asymmetry is particularly pronounced in Features 2, 6, 7, and 10, where the data distributions are notably skewed toward higher values.

Specific instances that exceed those thresholds represent univariate outliers. Within this dataset, Features 3 and 4 contain three instances categorized as outliers due to their unusually low values. Conversely, all other identified outliers across the remaining features are characterized by values that exceed the upper fence of the respective distribution.

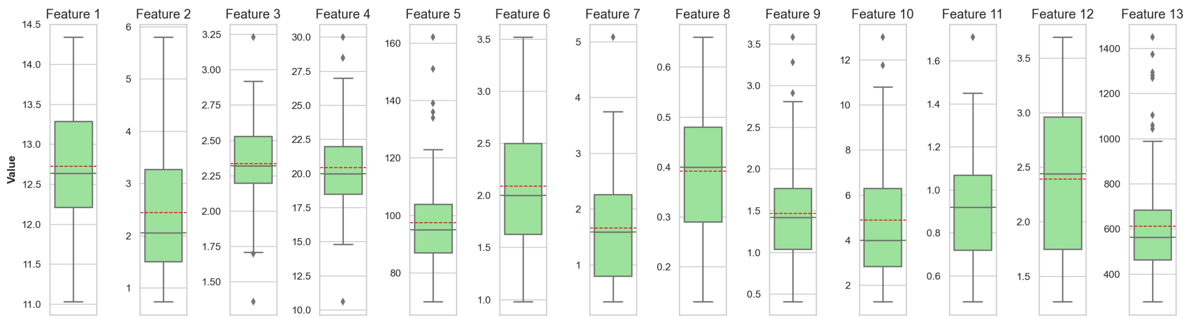


Figure 10: Boxplots per Dimension.

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The correctly identified outliers are highlighted in red. While the boxplot analysis captures 8 out of 10 ground-truth outliers, it also yields 18 false positives. These eight correctly identified instances are characterized by extreme values within feature 13, suggesting that the majority of the outlier class possesses an unusually high concentration of this component. However, such extreme values may not be inherently abnormal in a multivariate context, as their positioning could be explained by correlations with other dimensions.

To further investigate the distributional characteristics, Q-Q plots (Wilk and Gnanadesikan, 1968) for all features are provided in the Appendix. Additionally, the Shapiro-

Wilk test (Shapiro and Wilk, 1965, p. 592) was applied to each feature. The Shapiro-Wilk test evaluates the null hypothesis that a dataset is normally distributed and is specifically designed for continuous variables. In their seminal work, Shapiro and Wilk originally developed the test for relatively small sample sizes, specifically where $n < 50$. However, subsequent mathematical refinements by Royston (1982) extended the applicability of the calculation to larger datasets, allowing for the analysis of up to $n < 2000$ observations. Given these parameters, the requirements for the application of the Shapiro-Wilk test are satisfied in the present context.

Of the 13 analyzed dimensions, only Features 2 and 3 exhibited no significant deviation from a normal distribution, with recorded p-values of $p>0.05$ and $p>0.1$, respectively. Because the remaining features do not follow a normal distribution, the broader assumption of multivariate normality for the dataset must be rejected (Korkmaz et al., 2014, p. 156).

Bivariate Analysis

To further analyse the relationships in the dataset, a bivariate examination of the correlations between features is conducted. In figure 11, the Spearman rank correlation coefficients (Spearman, 1904, p. 80) for all pairs of dimensions are visualized using a heatmap. The Spearman correlation was chosen because it does not require normally distributed data, unlike the Pearson correlation Pearson (1896) Furthermore, it operates on the rank-order of values therefore the Spearman rank correlation is robust against outliers. This metric quantifies the strength and direction of the monotonic relationship between two variables, assigning a numerical value within the range of -1 to 1.

The correlation coefficients between the dimensions exhibit significant variation. A positive coefficient indicates that if the values of one feature increases the value of the other increases, whereas a negative coefficient implies an inverse relationship. A coefficient of zero suggests that there is no monotonic relationship between the variables.

The strongest positive correlation is observed between features 6 and 7, with a coefficient of ≈ 0.79. Similarly, a high positive correlation of ≈ 0.77 exists between features 7 and 12. Regarding inverse relationships, feature 11 and feature 10 exhibit a strong negative correlation of ≈ -0.61. The second most pronounced negative correlation, at approximately ≈ -0.60, occurs between Features 12 and feature 10 and feature 2 and 11.

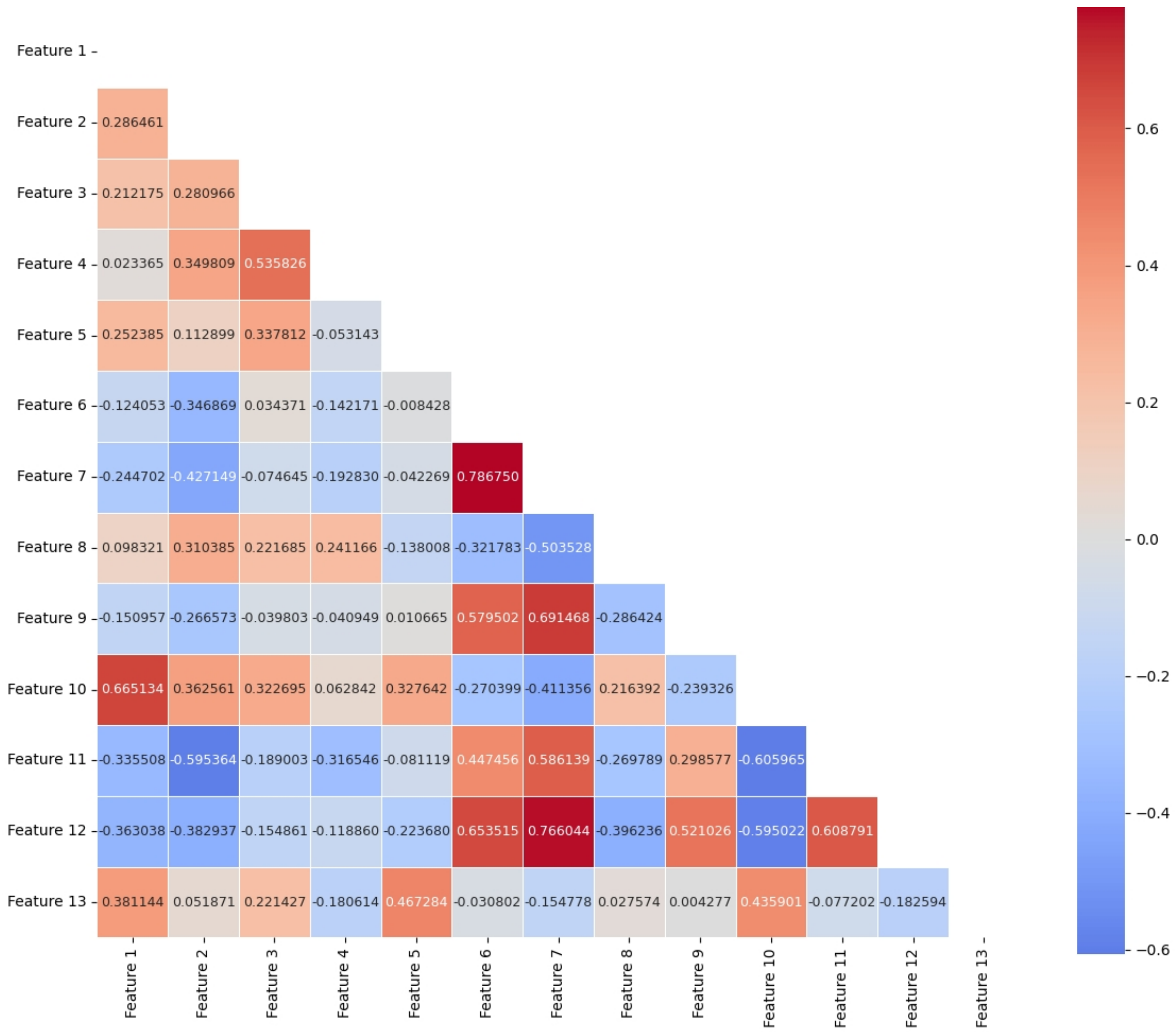


Figure 11: Heatmap of Spearman Correlations.

Manifold Structure Analysis

Analyzing the correlational structure of the dataset provides bivariate insights, yet it fails to capture the comprehensive geometric manifold of the data. To achieve a more profound understanding of the underlying data topology, Uniform Manifold Approximation and Projection (UMAP) (McInnes et al., 2018) is employed as a non-linear dimensionality reduction technique to visualize the high-dimensional observations in a lower-dimensional space. The subsequent section delineates the mathematical foundations of the UMAP algorithm, followed by an examination of the hyperparameters that govern the embedding process and a justification for the specific configurations selected for this study. Finally, the resulting visualization is presented alongside a detailed interpretation of the identified clusters and structural patterns.

The geometric manifold in the p-dimensional space can be highly complex globally, but locally follow a simple Euclidean structure. The goal of UMAP is to reduce the dimensional space and to display a visualization in two-dimensional space, while preserving the global and local geometric structure.(McInnes et al., 2018)

UMAP requires the assumption that the data is uniformly distributed on the manifold structure, which allows it to estimate the local connectivity of the data points. Without this assumption, the visualization would be dominated by density rather than the dataset's structure.

UMAP constructs the low dimensional depiction in two steps:

1. Construction of the high-dimensional graph by determining the local connectivity of each instance:
    - Estimation of distances between x_i and its k nearest neighbors.
    - Calculation of ρ as the distance to the nearest neighbor. By subtracting ρ from all neighbor distances, UMAP ensures that every instance maintains at least one strong connection, preserving the local connectivity.
    - Adaptation to local density via σ_i, see equation 9. Resulting in a smaller σ_i in the locally dense regions, while a larger σ_i in sparse regions.
    - Construction of a fuzzy simplicial set to merge these local views into a consistent global topological structure. In this context, a fuzzy simplicial set can be intuitively understood as a weighted graph where the edge strengths represent the likelihood of a topological connection between instances
2. Generating the low-dimensional representation:
    - Instead of random placement, UMAP uses spectral embedding to provide a stable initial layout. This captures the global topology early on and prevents the optimization from converging into poor local minima.
    - The algorithm iteratively adjusts the 2D positions by minimizing the fuzzy set cross-entropy.
    - Using Stochastic Gradient Descent, points with high connectivity are pulled together (attraction), while unconnected points are pushed apart (repulsion). This continues until convergence is achieved.

(McInnes et al., 2018, p. 6)

Equation 9 describes how UMAP adapts the local connectivity and determines the parameter σ_i.

$$\sum_{j=1}^{k} \exp\left(\frac{-\max(0, d(x_i, x_j) - \rho_i)}{\sigma_i}\right) = \log_2(k). \tag{9}$$

Where

$$\exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right), \tag{10}$$

represents the weight of the connection between instance x_i and its neighbor x_j. By subtracting ρ_i UMAP ensures that the closest neighbor always maintains the strongest connection, as $\exp(0) = 1$. By applying the exponential function, UMAP ensures that the influence of neighbors decreases with distance, creating a smooth decay in connectivity. σ_i has to be set that the sum of the exponential function over all neighbors equals $\log_2(k)$. $\log_2(k)$ was chosen, because it ensures that the entropy of all neighbors are consistent. A numerical optimzation process, known as binary search is applyied to find the appropriate σ_i for each instance.

To meet this requirement, UMAP assigns in locally dense regions a smaller σ_i, which effectively stretches the space. Conversely, in sparse regions, a larger σ_i compresses the space. This ensures the assumption of uniform distribution is met.

On the one hand, that depicts an outlier as an isolated instance apart from the main cluster, which coincides with the global outlier definition. On the other hand, the outliers is still bound to the manifold through the nearest neighbor. Therefore the absolute distance in the diagram loses its meaning.

When UMAP generates the low-dimensional representation, it employs a force-directed layout approach based on attraction and repulsion. Attraction occurs between points that exhibit strong connectivity in the high-dimensional space. Conversely, for instances with weak or non-existent connections, repulsion forces them apart in the two-dimensional embedding. These forces are adaptive to the previously calculated local affinities. This optimization results in a layout where clusters are grouped together even if they are sparsely distributed in the high-dimensional space, while isolated instances, often representing global outliers,are pushed away from the main data structures.

Hyperparameters

The configuration of UMAP necessitates the specification of two primary hyperparameters (McInnes et al., 2018, p. 23)that govern the construction of the low-dimensional embedding. Because these parameters determine the balance between preserving local and global structures, a universally optimal setting does not exist. Instead, the selection depends on the specific analytical objectives.

The first hyperparameter, $n_{neighbors}$, defines the size of the local neighborhood considered when approximating the manifold structure. This parameter manages a trade-off between local and global features. A low $n_{neighbors}$ value constrains the algorithm to focus on fine-grained patterns and intricate manifold details, which facilitates the identification of small clusters. Conversely, a high value incorporates a larger number of neighboring points, thereby prioritizing the broader relationships between distant regions of the data space. Selecting an appropriate value is essential for determining whether the data behaves as a continuous manifold or a series of discrete clusters.

The second hyperparameter, min_{dist}, establishes the minimum allowed distance between points in the low-dimensional embedding. This parameter controls the density of the visualization by determining how compactly points are packed. A smaller minimum distance allows for the representation of dense clusters, whereas a larger distance forces a greater separation, emphasizing the broad topological layout over local density.

To generate the visualization, the default hyperparameters $n_{neighbors} = 15$ and $min_{dist} = 0.1$ were utilized. The primary objective was to evaluate whether the Wine dataset exhibits distinct clusters or if the classes are topologically integrated. These standard settings provide a balanced representation that preserves both the local connectivity required for group identification and the global structure needed to observe transitions between regions. This configuration enables an objective assessment of the separability between the anomaly class and the inlier distribution in a two-dimensional environment. UMAP employs a stochastic optimization process, meaning specific coordinates may vary across different executions. However, iterative adjustments indicated that the underlying cluster formations remain consistent. Supporting evidence for this stability is provided in the appendix, where the algorithm was evaluated with $n_{\text{neighbors}} \in \{10, 20\}$ and $min_{dist} = 0.5$. To ensure all attributes contribute proportionally to the manifold construction, the data is normalized using the Robust transformation.

The axes in the resulting UMAP plot are not directly interpretable because UMAP is a non-linear dimensionality reduction technique that projects high-dimensional data into an abstract manifold. In this space, the coordinates do not correspond to specific physical units or original features because the algorithm prioritizes preserving topological relationships, such as neighborhood connectivity, over the linear distances or the variance of specific dimensions. Consequently, while the absolute coordinates are arbitrary, the spatial proximity between points serves as a proxy for the similarity of instances in the original high-dimensional space, where smaller distances indicate greater characteristic resemblance.

In contrast to alternative techniques, UMAP is designed to preserve a higher degree of global structure. Nevertheless, quantitative measurements of distances between clusters should be interpreted with caution, as the non-linear transformation into a lower-dimensional space does not maintain absolute linear distances from the original manifold.

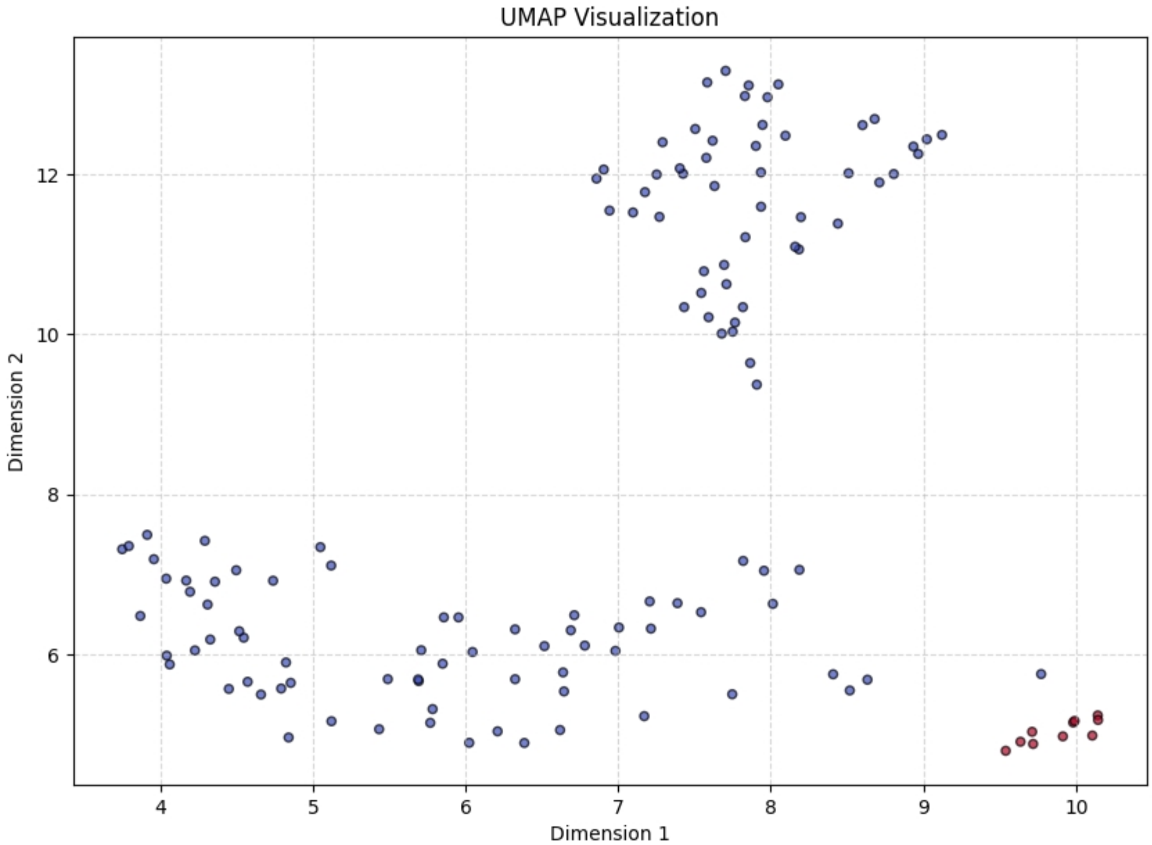


Figure 12: UMAP of the Wine Dataset.

The observations are color-coded according to their ground truth labels, with red indicating the outlier class and blue representing the inlier class. The resulting visualization reveals three distinct topological formations and is displayed in figure 12.

The inlier class appears as a relatively cohesive yet dispersed distribution partitioned into two primary sub-clusters, whereas the outlier class forms a highly concentrated and isolated cluster. Within the inlier class, the proximity of the observations suggests further distinct sub-clusters, including an elongated manifold and a dense upper-center grouping, which indicates a heterogeneous manifold for the normal data. Conversely, the clear spatial separation and high density of the outlier cluster, situated at the extreme right of the projection, indicate that these instances are distinct from the majority of the data. A single inlier observation is positioned in relative proximity to the outlier cluster, suggesting a potential boundary case or a marginal increase in similarity between the classes at that specific coordinate. This geometric isolation suggests that the profiles of the outliers are significantly and consistently different from those of the inliers.

In summary, the Wine dataset comprises 13 dimensions, 11 of which exhibit non-normal univariate distributions. Consequently, the multivariate distribution deviates from normality, which contradicts the underlying assumptions of the Minimum Covariance Determinant (MCD) estimator as detailed in Section 3.2. The correlation analysis reveals diverse directional relationships, including several strong dependencies between variables. While UMAP projections illustrate three distinct clusters—one of which represents the outlier class—certain instances of the normal class reside in close proximity to the outlier boundaries.

4.2 Metrics

Performance metrics are indispensable for evaluating the efficacy of outlier detection algorithms and facilitating rigorous comparative analyses between disparate methodologies. This selection process necessitates a balance between utilizing metrics widely established within the scientific community to maintain benchmarking standards and adopting metrics specifically suited to the inherent challenges of outlier detection, such as class imbalance. Consequently, this section focuses primarily on metrics derived from the confusion matrix.

The Area Under the Receiver Operating Characteristic curve (ROC AUC) and the Area Under the Precision-Recall Curve (AUPRC) are employed to determine the performance of the investigated methods, as these metrics are threshold-independent and provide a comprehensive evaluation of classification performance across various operating points. Furthermore, the F1-score is included to ensure the reproducibility of the results presented by Shenkar and Wolf (2022). To establish the necessary theoretical foundations, the fundamental components of these metrics, specifically Precision, Recall, and Specificity, are defined.

Although the evaluation utilizes benchmark datasets, the primary objective is to assess the methods within the context of real-world scenarios. Computational performance indicators, such as CPU utilization, execution time, and specific training or testing durations, are excluded from the scope of this research, as the analysis centers on the discriminatory power of the algorithms rather than their computational efficiency.

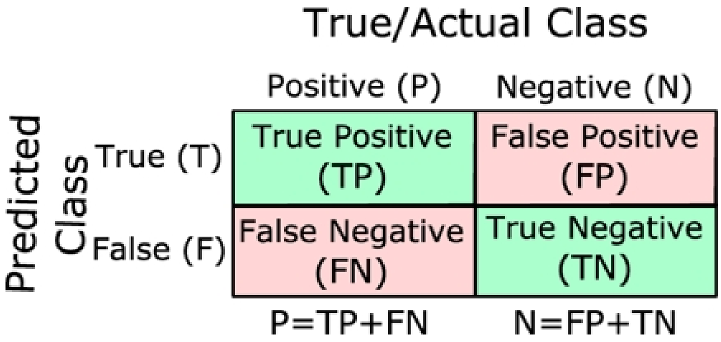
Within the context of outlier detection research, where benchmark datasets are available, the known outlier ratio is frequently utilized as the classification threshold to prevent the selection of arbitrary values. Consequently, threshold-dependent metrics such as Accuracy, Precision, and the F-Score are common in the scientific literature. An analysis by Nassif et al. (2021) of 290 research articles regarding outlier detection found that 112 papers employed accuracy, 54 utilized Precision, and 51 used F-Score metrics (Nassif et al., 2021, p. 78664). A significant finding of this literature review is that 64 of these 290 papers relied exclusively on a single metric, typically Accuracy or ROC AUC (Nassif et al., 2021, p. 78664). Nevertheless, Fawcett (2006) argue that threshold-dependent metrics are inferior to threshold-agnostic metrics because the selection of a specific cutoff remains arbitrary in real-world scenarios (Fawcett, 2006, p. 864). A threshold set excessively high increases the incidence of false negatives, whereas a threshold set too low elevates the number of false positives. Depending on the metric employed, such variations can produce deceptively high performance scores and distort the interpretation of the results (Fawcett, 2006, p. 4). Due its limitations, particularly in the presence of class imbalance, Accuracy is excluded from the performance evaluation in this thesis Valverde-Albacete and Peláez-Moreno (2014).

Confusion Matrix

In the context of outlier detection, performance is evaluated using a 2×2 confusion matrix (Tharwat, 2021, p. 170). This matrix facilitates a direct comparison between the predicted classifications and the actual ground truth labels. The structure of the confusion matrix is organized such that the horizontal axis represents the predicted classes generated by the detection method, while the vertical axis represents the true classes.

As seen in figure 13 the observations are assigned to one of the four cells: True positives (TP) and true negatives (TN) represent correct identifications of outliers and inliers, respectively, whereas false positives (FP) and false negatives (FN) represent misclassification errors. This systematic categorization serves as the foundation for calculating Precision, Recall and Specificity.

Precision quantifies the proportion of correctly identified outliers relative to the total number of instances classified as positive, encompassing both true positives and false positives. This metric emphasizes the degree of agreement between the ground truth



Source: (Tharwat, 2021, Fig. 1)

Figure 13: Confusion Matrix.

labels and the positive predictions generated by the classifier. It is formally defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{11}$$

Recall measures the proportion of correctly identified outliers relative to the total number of actual anomalies in the dataset. This metric effectively evaluates the model's ability to capture the entire anomaly class. It is formally defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{12}$$

Specificity quantifies the proportion of correctly identified inliers relative to the total number of actual negative instances in the dataset. This metric focuses exclusively on the negative class, serving to evaluate the effectiveness of a method in correctly rejecting normal observations. It is formally defined as:

$$\text{Specificity} = \frac{TN}{FP + TN} \tag{13}$$

Area under the ROC-Curve

ROC AUC serves as an aggregate performance metric evaluating a model with values constrained between 0 and 1. Those values represents the probability that a randomly selected outlier receives a higher anomaly score than a randomly selected inlier. Consequently, higher values indicate a superior capability of the model to distinguish between these two classes. The ROC curve plots the Recall on the ordinate against the False Positive Rate (1-Specificity) on the abscissa. For a graphical depiction figure 15 presents the resulting ROC AUC and AUPRC curves of the experiments. The area under this curve is mathematically defined by the integral.

Visually, a higher ROC AUC corresponds to a curve that bends toward the coordinate (0,1), where $ROCAUC = 1$ signifies a perfect classification with maximal False Positive Rate and Recall. A value of $ROCAUC = 0.5$ indicates that the model's predictive performance is equivalent to random chance, which establishes a baseline for comparative analysis and is visually represented by a diagonal line. Conversely, $ROCAUC = 0$ indicates that the model perfectly misclassifies outliers as inliers and vice versa, resulting in a curve that bends toward the coordinate (1,0).

The ROC AUC has a distinct disadvantage in imbalanced datasets, because it can yield overly optimistic assessments by incorporates False Positive Rate. In outlier detection the number of true negative instances typically far exceeds the number of true positives, the denominator of the False Positive Rate becomes dominated by true negatives (Davis and Goadrich, 2006). Consequently, the shift in the False Positive Rate resulting from the misclassification of inliers remains marginal. In imbalanced datasets, the False Positive Rate tends to remain low, which biases the curve toward the upper-left coordinate and may obscure poor model performance. A key strength of ROC AUC is its consistency across varying outlier ratios, which facilitates reliable comparisons between different models.

The AUPRC addresses the tendency of ROC AUC to provide overly optimistic performance assessments by excluding true negatives from its calculation (Saito and Rehmsmeier, 2015, p. 12), thereby focusing on the primary objective of detecting true positive instances.

Precision and Recall exhibit an inverse trade-off relationship, hence this represents the inherent tension between identifying every outlier (Recall) and ensuring those identified are correctly classified (Precision).

Area under the Precision-Recall Curve

The AUPRC (Davis and Goadrich, 2006) is bounded between 0 and 1, where a value of 1 signifies a perfect model that correctly identifies all outliers without false positives. The AUPRC plots Recall on the abscissa against Precision on the ordinate, with superior model performance indicated by a curve that extends toward the upper-right coordinate. It is formally defined as the integral of the precision-recall curve, which is calculated by varying the classification threshold across its entire range.

The curve generally declines as the classification threshold is lowered and outliers are ranked. At strict thresholds, only the most definitive outliers are identified, resulting in high precision but low recall. Conversely, lowering the threshold increases recall but also the likelihood of misclassification, which reduces precision and causes the curve to trend toward the lower-right region of the diagram.

Unlike the ROC AUC, the AUPRC does not possess a fixed baseline. Instead, the baseline is determined by the specific dataset and corresponds to the outlier ratio. In the case of a naive model with no discriminative power, the probability of such a model identifying a true positive is equal to the prevalence of outliers within the dataset. As the classification threshold becomes more lenient, the model identifies a greater number of true positives, thereby increasing recall. However, precision remains constant because the proportion of outliers is fixed. The increase in true positives occurs proportionally to the increase in false positives. Consequently, the baseline for a non-discriminative model is represented by a horizontal line at the level of the outlier ratio (Flach and Kull, 2015, p. 3).

This framework allows for the performance comparison of different methods in a given dataset: if the curve of Model A consistently remains above that of Model B, Model A demonstrates superior performance. If the curves intersect, one model may be better suited for tasks requiring high precision, while the other excels in high-recall applications. If a model's curve coincides with the baseline, its performance is equivalent to random guessing.

F1 Score

The F1-score (Tharwat, 2021, p. 174) represents the harmonic mean of precision and recall. It is formally defined as:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

The score is significantly influenced by the lower value. Consequently, a high F1-score is only achievable when both precision and recall are simultaneously high. Adjustments to the classification threshold increases typically one value and decreases the other. By optimizing the F1-score, the model effectively seeks a balance that minimizes both false positives and false negatives, making it a robust evaluation tool for imbalanced outlier detection tasks.

A primary critique of the F1-score is its exclusion of true negatives. Furthermore, the F1-Score assigns equal weight to Precision and Recall. Depending on the specific domain of the analysis, this can be addressed by reweighting the importance of Recall or Precision. This leads to the F_β family of metrics, such as the F2-score, which prioritizes recall, or the F0.5-score, which places greater emphasis on precision.

In conclusion, the AUPRC serves as the primary metric for this thesis, as it provides a more nuanced interpretation of the outlier class. By concentrating on the successful detection of anomalies while minimizing the incidence of false positives, the AUPRC offers a comprehensive assessment of algorithmic performance, particularly in the presence of class imbalance. The ROC AUC evaluates the efficacy of prediction ranking rather than absolute classification values, establishing a robust baseline against a random classifier that facilitates standardized comparative analyses. Furthermore, the F1-Score is included to maintain reproduce the results of Shenkar and Wolf (2022), specifically to allow for a direct comparison with performance of the k-nearest neighbors algorithm.

4.3 Implementation and Experimental Setup

This chapter delineates the technical specifications and configurations utilized throughout the experimental phase of this research. The following sections provide a comprehensive description of the hardware environment and the software architecture to facilitate the exact reproduction of the reported results. Thereafter the implementation of InterCont will be described, while explaining the self-programmed loop.py. The implementation of the benmark methods will be described thereafter.

The experiments were conducted on a workstation equipped with an Intel Core i9-12900K processor operating at 3.2 GHz and an NVIDIA GeForce RTX 3090 GPU. The system features 64 GB of physical RAM and 24 GB of video memory (VRAM), providing sufficient computational capacity. The software environment was managed under the Windows 11 Enterprise operating system, specifically version 25H2.

For the implementation of the algorithms and the execution of the experimental pipeline, the PyCharm 2025.3.3 IDE was employed. Although more recent versions of the Python interpreter exist, Python 3.12 (Python Software Foundation, 2023) was selected as the foundational language to ensure full compatibility with the required library ecosystem, as several critical packages had not yet been updated for newer releases. To maintain strict control over software dependencies and ensure environment stability, a virtual environment was utilized. All specific library versions and dependencies are documented within a provided requirements.txt file.

InterCont

The authors of InterCont provided a comprehensive reproduction package alongside their publication, which enabled a seamless replication of the original results. However, the provided codebase did not include implementations for k-NN. Consequently, k-NN and MCD were independently implemented for this study.

The reproduction package is structured into four distinct Python modules: main.py, train.py, data_loader.py, and helper_functions.py. The main.py script serves as the primary orchestrator for the experimental pipeline, managing configuration parameters, initiating data loading procedures, and executing the training sequence. The adjustable parameters include the dataset selection, an option for an accelerated version of the algorithm, and the batch size. The default batch size is set to 3000, which, given the dimensions of the datasets utilized in this research, effectively results in a full-batch gradient descent. This parameter was adjusted to 129, to ensure full batch learning.

The dataset utilized for the experiments can be specified via command-line arguments. However, since the provided reproduction package includes the entire ODDS repository and the default value selected the Wine dataset, this parameter remained unchanged. Furthermore, the configuration includes an option to enable an accelerated version of the algorithm. This feature was not activated, as the primary dataset is of a manageable size that does not necessitate the optimizations offered by the fast implementation. To ensure computational efficiency, the authors included a logic sequence at the conclusion of the execution script to prioritize the use of the GPU.

At the initialization of this module, a data pipeline is established through a specialized data wrapper, which can be found in the data_loader.py. For this implementation the function def build_train_test_generic_matfile is used. This component determines the structural dimensions of the dataset and defines the for individual row retrieval.

This function is essential for the implementation of the benchmark methods. To facilitate the subsequent evaluation of calculated anomaly scores, the wrapper preserves the original index identifiers. This architectural choice ensures that the machine learning algorithm can accurately map each anomaly score back to its corresponding observation in the source data.

To introduce stochasticity into the shuffle process, the authors utilized the PyTorch (Paszke et al., 2019) function torch.randperm(). This function divides the normal data into halves and shuffles the compositions and grading those halves into testing and training dataset. The testing dataset always includes all outliers. To ensure the stability and reliability of the results, a shuffling procedure is integrated into the pipeline. The torch.randperm() function relies on a pseudo-random number generator. Since no specific generator argument was specified, the default engine was employed.

As the reproduction package did not include a native looping mechanism for these iterations, a custom loop.py was implemented to manage the process. Finally, performance metrics were derived in accordance with the logic defined within the helper_functions module. Consistent with the original study, the experiments were executed over 500 random splits.

While the original codebase was designed to accommodate a vast array of datasets, including specialized logic within the data loader for .mat files, many components proved redundant for the specific dataset analyzed in this research. This code was retained to preserve the integrity of the reproduction package, as redundant logic does not influence the experimental outcomes. Notably, several hyperparameters within the original implementation were optimized for larger, higher-dimensional datasets. With exception of the bath size, none of the hyperparameters were changed. Furthermore, while the original authors restricted their evaluation to F1-Score and ROC AUC metrics, this study extends the analysis to include AUPRC. Minor adjustments were made to the source code to facilitate this additional metric.

k-NN

The k-NN algorithm was implemented in knn_benchmark.py by using the PyOD library (Zhao et al., 2019). To facilitate the experimental procedure and derive relevant performance metrics, components from the Scikit-learn library (Pedregosa et al., 2011) were also employed. Especially the F1-Score provided by Scikit-learn was used to estimate this metric. Nevertheless, by using the known outlier score, the use of both methods yield to the same result.

A five-fold cross-validation procedure was implemented in knn_cv.py. This was facilitated by the KFold function from the Scikit-learn library. $K = 5$ is the optimal value for k, which corrosponds with the choice in the paper of the authors. Given the limited number of observations in the dataset, a five-fold approach was selected to maintain a sufficient volume of data in both the training and testing subsets for each iteration. The shuffle parameter *random_state* was enabled to ensure that the observations were randomized prior to partitioning and was set to the default 42.

To ensure a rigorous and consistent evaluation, the same framework utilizing the torch.randperm

() function, was integrated into the k-NN implementation. In this configuration, the k-NN algorithm establishes a representation based on the normal observations within the training set. Subsequently, the method calculates the distances between each test observation and its k nearest neighbors from the training data. The anomaly score for each test point is derived from the maximum distance among these k nearest neighbors. The dataset includes 119 normal datapoints, which can not be seperated equally into training and testing. To align the model with the specific characteristics of the testdataset, the number of outliers are hardcoded into the script.

The k-NN implementation utilizes the default configuration for distance metrics, specifically selecting the largest distance to the k-nearest neighbors. This calculation employs the Minkowski distance with the parameter $p = 2$, which is mathematically equivalent to the Euclidean distance. The RobustScaler from Scikit-learn was implemented.

FAST-MCD

The FAST-MCD algorithm is implemented in mcd_benchmark.py utilizing the PyOD library. As an unsupervised method, FAST-MCD does not support a traditional separation into training and testing phases. If a split-sample approach were attempted, the algorithm would collapse. Specifically, if $n = h$, the resulting $\mu_M CD$ and $\sum_M CD$ matrix would be equivalent to the mean and covariance matrix of the training dataset (Rousseeuw and Driessen, 1999, p. 217). To ensure experimental comparability, the FAST-MCD algorithm is applied directly to the test dataset. The training dataset is generated using the same shuffle function employed by the other evaluated methods. Observations that exhibit the 10 highest Mahalanobis distances relative to the estimated mean and covariance matrix are classified as outliers, adhering to a top_k=10 criterion.

To ensure parity with the implementation of InterCont, this cycle was repeated 500 times. The final performance metrics represent the mean values across these iterations, supplemented by the calculation of the standard deviation. The random_state parameter was fixed at 42 to guarantee reproducibility, because FAST-MCD relies on selecting random initial subsets before performing subsequent C-steps.

5 Results

This section presents the results of the previously conducted experiments. The primary findings are summarized in Table 1, which details the mean and standard deviation

(SD) of the F1-Score, ROC AUC, and AUPRC for each evaluated method. To ensure a rigorous assessment of these findings, a Brunne Munzel test (Friedman, 1937) is applied. The analysis is further extended through the plotting of AUPRC and ROC AUC curves to facilitate a more detailed performance comparison across all methods.

Table 1: Experimental Results of Benchmark Protocol

Method	F1-Score	SD	ROC AUC	SD	AUPRC	SD
MCD	2.8	4.14	17.15	6.54	9.72	4.53
k-NN	70.34	9.81	95.80	1.81	73.06	11.72
InterCont	**85.64**	8.77	**97.97**	2.13	**92.93**	7.31

In the F1-Score analysis, InterCont yields the highest mean at 85.64, followed by k-NN at 70.34 and MCD at 2.8. The SD values indicate that MCD maintains the lowest variability at 4.14, while InterCont and k-NN show higher fluctuations at 8.77 and 9.81.

The ROC AUC results show high discriminative performance across all methods, with InterCont reaching 97.97, k-NN 95.80 and MCD at 17.15. The stability of these scores is evidenced by low SD values, with k-NN demonstrating the least variance at 1.81, compared to 2.13 for InterCont and 6.54 for MCD.

For the AUPRC metric, InterCont achieves a mean of 92.93, significantly higher than k-NN at 73.06. MCD yields the lowest AUPRC at 9.72 and the lowest SD at 4.53. The InterCont method exhibits a SD with an SD of 7.31. Conversely, k-NN records a SD of 11.72.

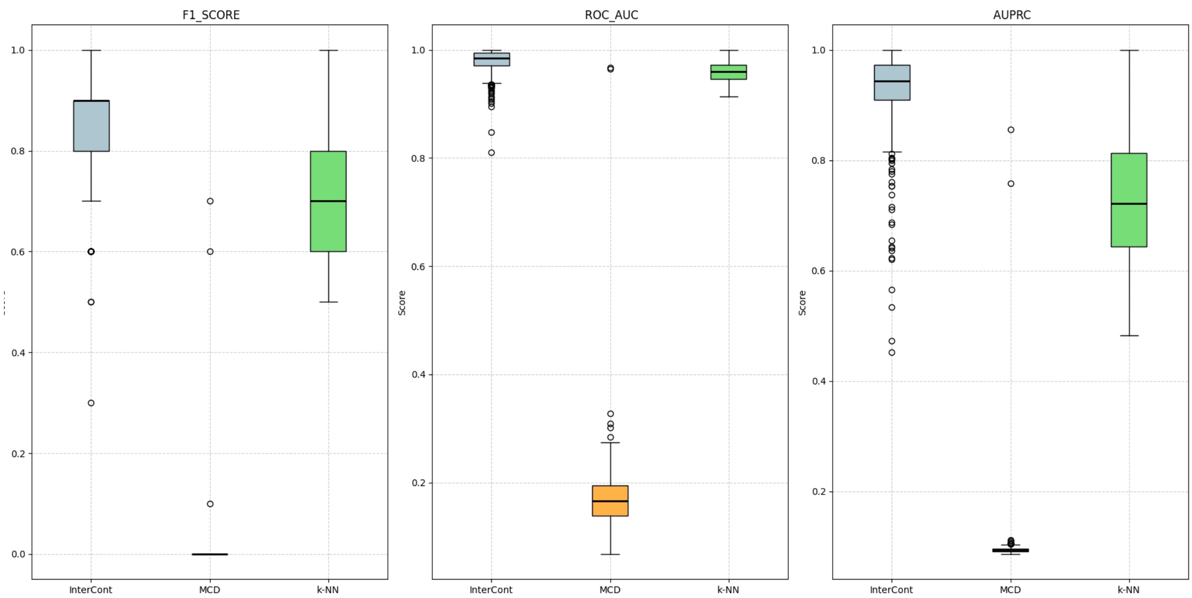


Figure 14: Boxplot of Experimental Results.

The figure 14 displays boxplots for the resulting metrics for each method. Regarding the ROC AUC metric, InterCont demonstrates the highest median performance, closely approaching a value of 1.0, while maintaining a relatively narrow interquartile range despite several lower-reaching outliers. K-NN exhibits a lower median ROC AUC value at $\approx$ 0.75. MCD displaying notable lower median ROC AUC above 0.2 and higher variance. Notably one single outlier reaches a value above 0.9.

In the AUPRC evaluation, InterCont consistently maintains a superior median position above 0.90, whereas the performance of k-NN declines significantly. The InterCont boxplot is the most compact, despite a high number of outliers extending below the lower whisker. K-NN exhibits a wider interquartile range, with its whiskers extending from 1.0 at the maximum to $\approx$ 0.5. MCD demonstrates a substantially lower median AUPRC. The entire boxplot for this method does not exceed 0.1, with the exception of two outliers reaching values above $\approx$ 0.7.

The F1 Score results further distinguish the algorithms, as InterCont achieves a higher and more consistent score than its counterparts. The median F1 Score for InterCont is $\approx$ 0.9, characterized by a relatively small interquartile range compared to k-NN. The median F1 Score of k-NN represents the second highest value at $\approx$ 0.7, with whiskers extending from 1.0 down to $\approx$ 0.5. The performance of MCD is visually indistinguishable from zero. However, three outliers are displayed above the interquartile range, the highest of which reaches $\approx$ 0.7.

Across all three metrics, InterCont consistently outperforms MCD and k-NN in terms of both central tendency and overall stability, despite the presence of isolated performance drops indicated by the outlier points. These results suggest that InterCont provides the most robust classification performance for the underlying dataset, particularly when considering the Precision-Recall trade-off captured by the AUPRC.

To further analyse the figure 15 displayd the ROC AUC and AUPRC curves of the three methods.

The ROC AUC curves for InterCont and k-NN exhibit a rapid initial increase. The InterCont method maintains a steeper ascent until Recall reaches 0.8. While the growth rate for k-NN is initially less pronounced, it surpasses InterCont at $\approx$ 0.1 False Positive Rate (1-Specificity) and remains superior thereafter. Conversely, the MCD curve falls significantly below the diagonal dashed line, which represents the performance of a random classifier. This inverse curve indicates that the MCD model, under the current configuration, performs worse than chance. Furthermore, the MCD curve intersects with InterCont at a False Positive Rate of $\approx$ 0.75.

AUPRC demonstrates significant variance across the evaluated models. InterCont method

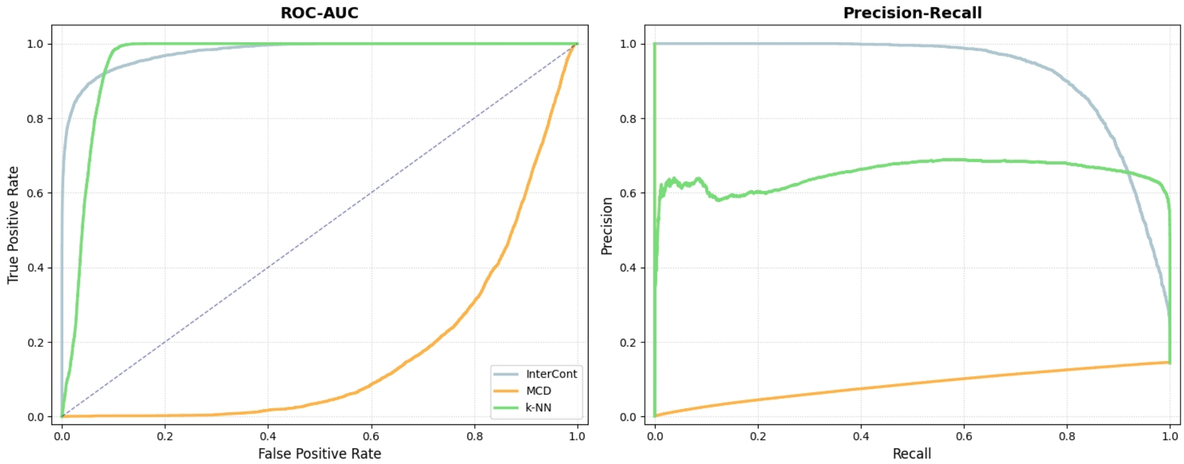


Figure 15: ROC AUC and AUPRC Curves.

exhibits the most stable and high precision across a wide range of recall values, maintaining a near-perfect precision until approximately 0.8 recall before declining. While InterCont maintains a consistently high performance over an extended range, k-NN exhibits a rapid decline in precision following an increases in recall. Eventually, k-NN reaches a plateau where precision stabilizes at ≈ 0.6. That indicates a consistent but higher rate of false positives compared to InterCont. At high values for recall, k-NN crosses the AUPRC curve of InterCont and falls steeply thereafter.

The MCD method displays very low precision that increases slightly as recall grows, which reinforces the observation from the ROC plot that this specific implementation fails to accurately distinguish the underlying outlier distribution in this context.

Statistical Inference

The validation of empirical results necessitates the application of rigorous statistical testing. To mitigate the distributional challenges identified, the non-parametric pairwise Brunner Munzel test (Brunner and Munzel, 2000) will be deployed. This procedure serves to determine whether significant differences exist across the three methods.

This non-parametric procedure is utilized within the paper of Shenkar and Wolf (2022).

The Brunner-Munzel test, also called generalized Wilcoxon test, is a test of the null hypothesis that, for two populations the porbability of one beging greater then the other equal as the other way around. This test is implemented in situations where the observations are independend. This assumption is met, because in the setup the shuffle function is excuted three different times, yielding different test set for all 500 runs.

Also this test is viable in situations where the variances of the populations differ. The standard deviations reported in Table 1 exhibit substantial variation, suggesting that the assumption of homoscedasticity may be violated. To formally evaluate the homogeneity of variances, Levene's test is employed to determine whether the variances across the different groups are statistically equivalent.

The null hypothesis H_0 for Levene's test posits that the population variances are equal across all groups, which can be formally expressed as:

$$H_0 = \sigma_1^2 = \sigma_2^2 = \sigma_3^2. \tag{15}$$

Levene's test also requires the assumption, that the observations are independend. This is satisfied as discussed above. Levene's test is robust against non-normally distributed populations, which is certainly not met seening in 14.

If the resulting p-value is below the chosen significance level, the null hypothesis is rejected, indicating that the variances are significantly different. The results are displayed in table 2. The results across all evaluated metrics demonstrate a significance level of $p < 0.001$, confirming that the requirement for homoscedasticity of variances has not been satisfied.

Levene's test and the boxplots in Figure 14 confirm substantial heteroscedasticity, the assumption of identical distribution shapes is not satisfied. Under these conditions, the Friedman test is interpreted as a measure of stochastic dominance rather than a direct comparison of medians. The results are displayed in table 2.

Table 2: Levene's Test for Homoscedasticity of Variances

Metric	**Statistic**	**p-value**
F1-Score	182.98	0.00
ROC AUC	69.15	0.00
AUPRC	286.93	0.00

The null hypothesis is rejected across all evaluated metrics, indicating that the variances exhibit statistically significant differences. To further delineate specific performance differentials, a pairwise Brunner-Munzel test is conducted. The null hypothesis for this test is defined as:

$$H_0^p : p = P(X_1 < X_2) + \frac{1}{2} P(X_1 = X_2) = \frac{1}{2}, \tag{16}$$

which posits that the probability of an observation from one distribution exceeding an observation from the other is equal to the reciprocal probability, signifying stochastic equality between the two samples.

To calculate the probabilities, the results from both methods are organized into a global ranking. In instances where identical values occur, the mean rank is assigned to the respective observations. Additionally, an intraclass ranking is established specifically for the method under observation. The estimator $\hat{p}$ is subsequently calculated as follows:

$$p = \frac{1}{n_1}(R_2 - \frac{n_2 + 1}{2}). \tag{17}$$

The median rank of the second method within the global ranking list is denoted by R_2. Given that n represents the same sample size, for this thesis $n_1 = n_2$ is satisfied. This estimation is derived from the global ranking. To assess whether the observed differences are equivalent, the discrepancy between the rank in the global ranking and the interclass ranking is calculated. Subsequently, the variance of this difference is analyzed for both methods.

The test statistic is estimated as follows:

$$W_N^{BF} = \frac{\bar{R}_{2\cdot} - \bar{R}_{1\cdot}}{\sqrt{N \cdot \hat{\sigma}_N^2}}. \tag{18}$$

This statistic quantifies the distance between the observed effect and the null hypothesis in units of standard deviations. The test statistic follows an asymptotic standard normal distribution.

As the analysis involves multiple pairwise comparisons between the three methods and three metrics, it is necessary to adjust the significance level. Without such an adjustment, the probability of encountering at least one Type I error across the set of tests increases, potentially leading to false-positive conclusions. To mitigate this, the Bonferroni correction is applied, which involves dividing the initial significance level ($\alpha = 0.05$) by the number of comparisons being performed (m=3). Consequently, for a result to be considered statistically significant and for the null hypothesis to be rejected, the calculated p-value must be lower than the adjusted threshold of 0.0167.

As discussed before, the normalization method applied in the k-NN could not be determined. Therefore second experiment with two different normalization methods will be employed to show if the result will reach the results of the paper of Shenkar and

Table 3: Results of the Brunner Munzel Test

Metric	F1-Score	ROC AUC	AUPRC
InterCont vs k-NN	$p < 8.40 \times 10^{-81}$	$p < 6.39 \times 10^{-31}$	$p < 6.11 \times 10^{-57}$
InterCont vs MCD	$p < 1.86 \times 10^{-62}$	$p < 2.36 \times 10^{-44}$	$p < 6.65 \times 10^{-73}$
k-NN vs MCD	$p < 1.81 \times 10^{-45}$	$p < 6.63 \times 10^{-1}$	$p < 1.25 \times 10^{-4}$

Wolf (2022) For that the same setup was used as before in the k-nn benchmark. The most commonly used normalization methods were applied: Min-Max-Normalization and Z-Score normalization (StandardScaler) The normalization settings where changed after performing k-NN in 500 runs for different values of k. Because thats a comparison of the paper, only the mean and SD of the F1 Score is displayed in the table 4. To test the statistical significance of the difference a friedman test was applied. There is a statistical significance between the result with $p < 0.001$ and a Friedman Statistics of 118.20, hence the depending on the normalization of the data the F1 Score differs.

Table 4: Results of the second Experiments

Metric	F1-Score	SD
Min-Max-Normalization	**76.36**	8.20
Z-Score Normalization	75.86	9.82

6 Discussion

This chapter synthesizes the empirical evidence with the theoretical frameworks of representation learning and robust statistics established in Chapters 2 and 3. The following section summarizes the primary findings.

The hypotheses formulated for this thesis are defined as follows:

- H1: k-NN will achieve the highest predictive performance across AUPRC.
- H2: While InterCont will outperform MCD, its AUPRC will remain secondary to k-NN.

Based on the results presented in Table 1, InterCont attains the highest mean AUPRC of 92.93, while MCD reaches a mean AUPRC of 77.82, k-NN reaches the lowest mean AUPRC of 73.06. The differences between InterCont and the benchmark methods are statistically significant at $p < 0.001$ within the Wilcoxon signed-rank test. The empirical evidence leads to the rejection of both H1 and H2. Contrary to the initial assumptions, the representation learning-based InterCont method did not remain secondary to the benchmarks but emerged as the superior approach, significantly outperforming both MCD and k-NN in all primary metrics. For this setup and the given dataset, the implementation of InterCont does provide a more presice outlier classification.

Figure 15 illustrates the ROC AUC curves for the three evaluated methods. Each curve converges toward the upper-right quadrant of the coordinate system, indicating that the models effectively distinguish between the respective classes. Furthermore, all methods demonstrate a high True Positive Rate while simultaneously maintaining a low false positive rate.

Initially, InterCont maintains a significantly higher Precision than the other methods, which demonstrates its robustness in identifying outliers under strict threshold conditions. In the ROC AUC plot, the curves of k-NN and MCD eventually cross and surpass InterCont. This indicates that k-NN and MCD achieve a higher True Positive Rate more quickly. However, the corresponding Precision-Recall curve reveals that this is achieved at the cost of a lower Precision due to increased false positives.

While k-NN appears superior in the ROC AUC metric, this result is somewhat misleading as the large number of normal instances masks the impact of false positives. The AUPRC confirms that InterCont is the more reliable model for precise anomaly detection, whereas k-NN is more effective at capturing a larger volume of anomalies if a lower Precision is acceptable.

The UMAP visualization in Figure 12 illustrates that the outliers in the Wine dataset form a concentrated and isolated cluster. With few exceptions, the outlier class remains distinct from the cohesive yet more dispersed inlier distribution. This topological formation suggests that the dataset primarily contains global and contextual anomalies rather than collective outliers. While a clear geometric separation generally favors distance-based approaches such as k-NN, these dataset characteristics may facilitate a more effective distinction for the InterCont algorithm. The application of MCD requires the assumption of normally distributed data, which was not satisfied, as depicted in Figure 16.

By utilizing a sliding window approach for data augmentation, InterCont effectively captures local feature dependencies in contrast to the global and distance-based perspectives of the benchmarks. Although the Wine dataset is characterized predominantly by monotonic relationships, this localized approach significantly improves the discriminatory power of the method. This is achieved through the InterCont architecture, which maps these local dependencies onto a larger feature space. This projection into a higher-dimensional space enables the method to distinguish even subtle irregularities that are non-separable in the original input space.

InterCont defines the training dataset's structure with greater granularity than the benchmark models. The superiority of InterCont is particularly evident in AUPRC. This discrepancy highlights a critical diagnostic reality: while the ROC AUC yields optimistic assessments by incorporating true negatives, the AUPRC reveals that the benchmarks are significantly less effective at identifying the specific minority class.

The integration of an ensemble strategy involving the permutation of instances ensures that the model systematically evaluates various feature orderings. This prevents critical relationships from being omitted due to arbitrary data structuring, ensuring the method captures all of feature interactions. Furthermore, the implementation of a double normalization process, applying both row-wise and column-wise L2 normalization, projects observations onto a unit hypersphere. This ensures that classification is based solely on the cosine similarity of the anchor with its positive and negative counterparts. The loss function maximizes the mutual information between positive pairs and minimizes it between negative pairs.

The k-NN algorithm yielded the lowest AUPRC of 73.06 and demonstrated performance equivalent to MCD in terms of ROC AUC. Figure 15 displays a sharp decrease in Precision at strict thresholds. This indicates that k-NN identified false positives among the instances with the greatest distances, suggesting that true negatives are also scattered within low-density regions of the dataset. Following this, precision increases, which indicates that a significant portion of the true positives is concentrated at lower score values. The k-NN algorithm successfully identifies more true positives than false positives in this range, resulting in an upward trend in the Precision-Recall curve.

Subsequently, the curve stabilizes above a Precision of 0.6, which indicates that true positives and false positives are detected in equal proportions. To achieve any further increase in Precision, the k-NN algorithm identifies an increased number of false positives. Consequently, the Precision-Recall curve decreases as it proceeds further toward the left. The curse of dimensionality may have negatively impacted the performance of the k-NN algorithm. As dimensionality increases, the distances between neighbors tend to converge, which can lead to an increased misclassification of false positives and a subsequent reduction in the AUPRC.

This observation remains consistent across all 500 iterations of the normal data. Datasets possessing this specific structure prove disadvantageous for k-NN due to the underlying local density characteristics. Furthermore, the dataset contains noise, which includes significant feature overlap. Because k-NN relies solely on the local distances of its neighbors, critical information such as feature correlations is not incorporated into the model. Consequently, this approach exhibits less granularity in comparison to the InterCont algorithm.

The MCD estimator demonstrated the second highest performance among the methods. Even though the MCD estimation was done at the training dataset, the model fails Notably, the inherent affine equivariance of MCD, which theoretically eliminates the necessity for data normalization, provided no measurable advantage here. The superior performance of non-equivariant methods like InterCont and k-NN suggests that specific mechanics for handling feature interactions are more critical for detection efficacy than geometric stability. Furthermore, as evidenced by figure 16, the data is not Gaussian distributed, violating the fundamental distributional assumptions of the MCD estimator. Even when applying a semi-supervised approach, estimating the mean and covariance matrix from the normal training data, the estimator fails to achieve a clean separation. This instability is reflected in the high standard deviation of MCD's AUPRC results (SD=16.43), indicating that the robust covariance estimator was highly sensitive to the specific data splits.

The superiority of deep learning methods over traditional methods like k-NN aligns with the findings of Thimonier et al. (2023). In their study, they compared, among others, k-NN against their own NPT-AD method using the ODDS repository. The NPT-AD yielded an average F1-Score of 68.8 compared to 65.8 for k-NN. These results support the observations in this thesis, suggesting that neural architectures can capture more complex patterns than distance-based benchmarks.

The outstanding F1 Score of k-NN in the paper of Shenkar and Wolf was the main driver of this thesis. This performance could not be reproduced in this thesis. Table 5 compares the results of this thesis with the performance of the paper ones:

Table 5: Comparison of k-NN Performance

Metric	F1-Score	SD
This thesis	72.18	10.62
Shenkar & Wolf	94.00	4.9

Nevertheless the evaluation protocol has been followed, the F1 Score performance of k-NN in this thesis deviates from the performance of the paper. There is little information about the implementation of the benchmark k-NN in the paper of Shenkar Wolf, besides the evaluation protocol. It was not described which normalization method was used, hence a robustness check was executed and displayed in Table 4.

Table 4 presents the results of the Friedman test, indicating that the observed differences are statistically significant with $p<0.001$. These findings can be attributed to the specific distributional transformations imposed by each normalization technique. While the Robust Scaler is designed to preserve the underlying distribution by using median and interquartile ranges to mitigate the masking effect of outliers, Min-Max and Z-Score normalizations are more sensitive to extreme values.

In this specific setup, Min-Max normalization achieved the highest mean F1-score. This can be explained by the compression of the inlier data (Shrivastava and Vamsi, 2023). When global outliers occupy the extreme ends of the scale, the remaining normal observations are pushed into a very narrow range.

For a distance-based algorithm like k-NN, this extreme compression can be advantageous for identifying global outliers that deviate significantly from the dense cluster of inliers. However, such a transformation could be disadvantageous for datasets containing contextual or semantic anomalies, as these subtle irregularities might be compressed into the inlier distribution and become indistinguishable. Given that the dataset used in this study primarily contains global rather than contextual outliers, this specific disadvantage does not negatively impact the results.

The Z-Score normalization does not compress the data as Min-Max Normalization, but it is not robust against global outliers. It includes all instances for the normalisation not like the robust scaler. This could lead to higher F1 Score because the the information if the outliers is preserved, hence the recall increases and simultaneously constant Precision. In comparison the robust scaler could push instances outside the IQR further away increasing the noise and reducing the F1 Score.

However, these results do not achieve the performance levels reported by Shenkar and Wolf. Such discrepancies may arise from the specific experimental setup employed in this study. Although the experiments were implemented following the evaluation protocol described in the original paper, a semi-supervised approach was utilized across all methods to ensure comparability. For instance, the InterCont method first processes the normal data to abstract underlying characteristics before being applied to the abnormal dataset. A similar procedure was implemented for k-NN, where distances within the normal data are estimated prior to evaluating the abnormal dataset. Nevertheless, by permutating normal data into the test dataset, the varying degress of noise could explain difference in terms of F1 Score.

MCD is inherently a parameter estimation method. While it does not involve iterative learning in a connectionist sense, its performance relies on robustly estimating the location and scatter of the underlying distribution. These parameters were subsequently applied to the abnormal dataset, representing a specific configuration that is not present in other implementations of MCD.

Schindler et al. (2023) conducted experiments comparing unsupervised autoencoder networks against traditional machine learning methods like Isolation Forest (Liu et al., 2008). Their findings prove that autoencoder networks offer advantages in datasets that contain contextual outliers. Nevertheless, their methodology includes outlier injection, in which artificial outliers are injected into a dataset. This methodology necessitates a careful definition of the injected outliers, as the results depend on the decisions of the authors. In contrast, the present work utilizes the Wine dataset which avoids this potential bias.

In a recent paper, Shrivastava and Vamsi (2023) proposed a hybrid approach between machine learning and MCD. They used MCD in a first stage to determine Mahalanobis distances. Subsequently, they expanded the dataset with those distances, and the machine learning methods used this additional information to detect outliers. They found significant enhancements in F1-Score, Precision, and Recall for the methods that incorporated the robust Mahalanobis distances.

The application of MCD in this thesis follows the logic of hybrid approaches, such as the one proposed by Shrivastava and Vamsi (2023). While MCD is traditionally an unsupervised tool, its use as a feature generator or distance estimator for subsequent classification proves to be effective. The significant enhancements in F1-Score reported by Shrivastava and Vamsi (2023) support the decision to incorporate robust distance measures into the detection pipeline.

Criticism regarding the ODDS repository has emerged (Campos et al., 2016, p. 923), primarily because these datasets frequently contain easily identifiable global outliers rather than complex semantic anomalies. Furthermore, authors such as Han et al. (2022) have criticized the construction of the ODDS datasets. For instance, the Wine dataset was created by downsampling and repurposing existing classes to mimic anomalous behavior. Such artificial construction may not accurately reflect the challenges associated with detecting real-world outliers.

To standardize outlier detection procedures, Han et al. (2022) developed the ADBench framework. The authors evaluated 30 popular algorithms across 57 distinct datasets to test the models. By varying the learning paradigm, they investigated how the level of supervision affects the results, incorporating supervised, semi-supervised, and unsupervised approaches.

The authors categorized the data into local, global, and cluster anomalies to determine which methods perform optimally for each specific outlier type. To assess model robustness, the researchers implemented outlier injection to systematically increase the noise within a dataset, thereby facilitating a comprehensive robustness check. However, this procedure might bias results towards specific distribution assumptions.

The researchers observed that foundational algorithms frequently outperform complex representation learning methods, suggesting that the primary findings of their study support the hypothesis presented in this work. Although ADBench suggests that foundational algorithms often outperform complex models, the results in this thesis demonstrate that under the specific conditions of semi supervised learning, deep learning methods provide a clear advantage. This discrepancy highlights that there is no 'one-size-fits-all' solution in outlier detection.

7 Conclusion

This thesis conducted a comparative performance analysis of three distinct outlier detection methodologies: the robust statistical MCD using the FAST-MCD algorithm, the distance-based k-NN approach, and the representation learning-based InterCont method. The primary objective was to evaluate whether the increased algorithmic complexity of modern machine learning techniques translates into superior predictive efficacy compared to foundational statistical and distance-based benchmarks.

The results suggest that representation learning, specifically through the mechanisms of internal contrastive learning, sliding window data augmentation, and feature permutation, effectively captures complex dependencies within tabular data that traditional distance-based or distribution-assumed methods may overlook. While the MCD demonstrated high discriminative power as evidenced by its ROC AUC, its instability in AUPRC highlighted a sensitivity to specific data subsets and a reliance on Gaussian assumptions that are often violated in real-world datasets. Similarly, while k-NN provided an intuitive baseline, it proved susceptible to local noise and lacked the robust generalization capabilities offered by the asymmetrical neural network architecture of InterCont.

The empirical investigation utilized the Wine dataset from the ODDS repository, which was characterized by a significant class imbalance. Quantitative evaluation through 500 experimental runs demonstrated that InterCont achieved the highest predictive performance across all primary metrics, yielding a mean F1-score of 85.64, a ROC

AUC of 97.97, and an AUPRC of 92.93. Statistical inference using the Friedman test and pairwise Wilcoxon signed-rank tests confirmed that these performance gains were statistically significant with $p < 0.001$. Consequently, the research hypothesis H1 was rejected, which posited that k-NN would yield the highest performance. H2, asserting a higher AUPRC of InterCont over MCD benchmarks but leaving it behind k-NN, was rejected as well.

Despite the superior performance of InterCont, this study identified critical challenges regarding the reproducibility of results from high-complexity models and the inherent limitations of threshold-dependent metrics like the F1-score in imbalanced domains. Future research should aim to extend this unified benchmarking framework to a broader spectrum of real world high-dimensional datasets. In conclusion, this investigation affirms that for the specific task of identifying anomalies in multidimensional tabular datasets, Internal Contrastive Learning provides a significant and measurable advantage over Minimum Covariance Determant and the k-nearest neighbor algorithm.

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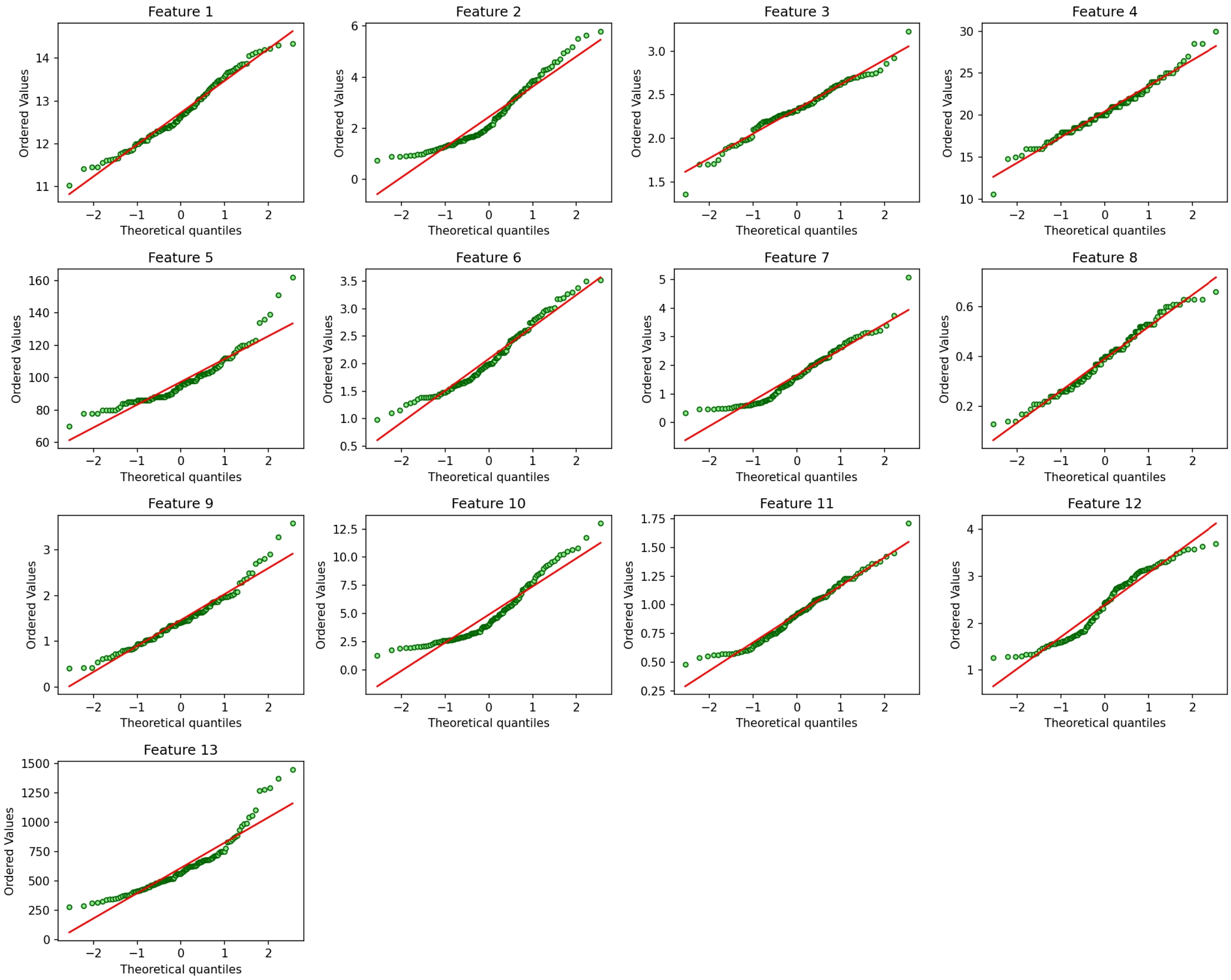

Figure 16: Q-Q Plot.

Table 6: Results of Shapiro-Wilk test

Method	Statistic	p-value
Feature 1	0.98	4.22
Feature 2	0.92	0.00
Feature 3	0.98	**7.98**
Feature 4	0.98	**10.08**
Feature 5	0.89	0.00
Feature 6	0.96	0.13
Feature 7	0.94	0.00
Feature 8	0.97	4.20
Feature 9	0.95	0.06
Feature 10	0.90	0.00
Feature 11	0.97	0.85
Feature 12	0.94	0.00
Feature 13	0.89	0.00

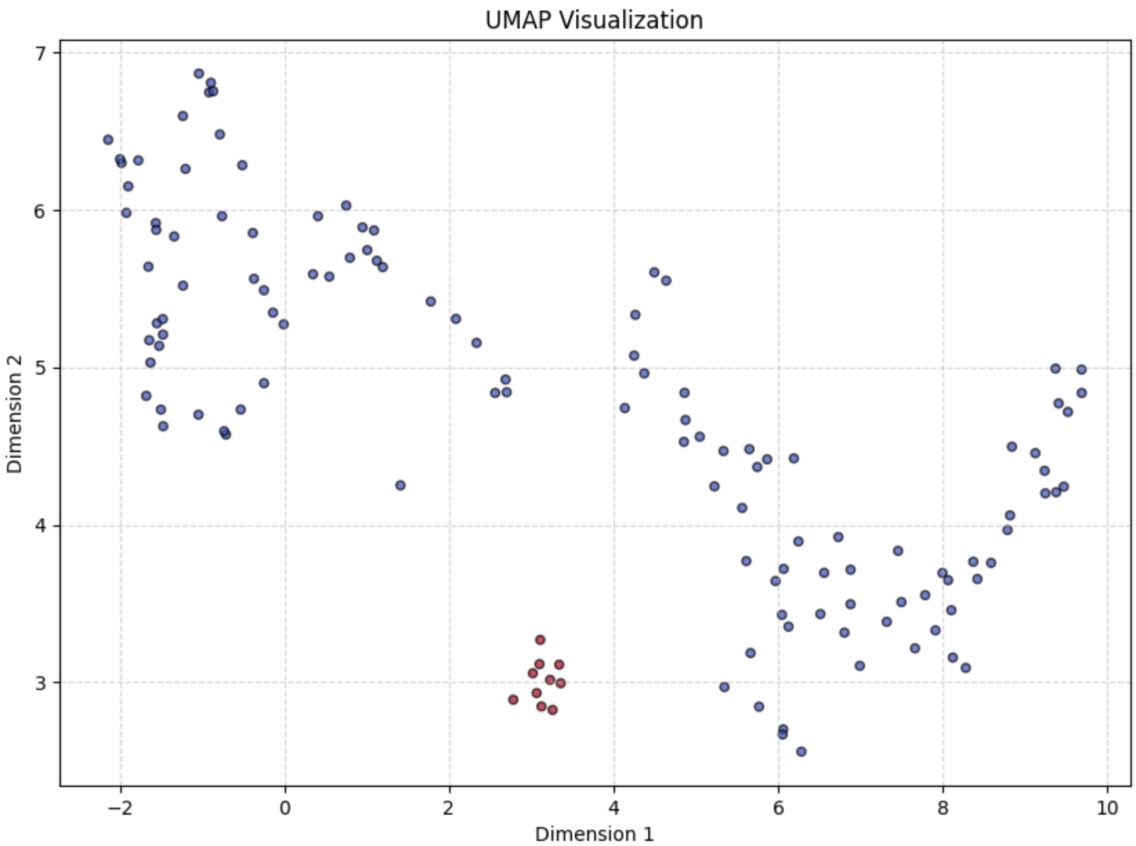

Figure 17: UMAP with $n_{neighbors} = 10$ and $min_{dist} = 0.1$.

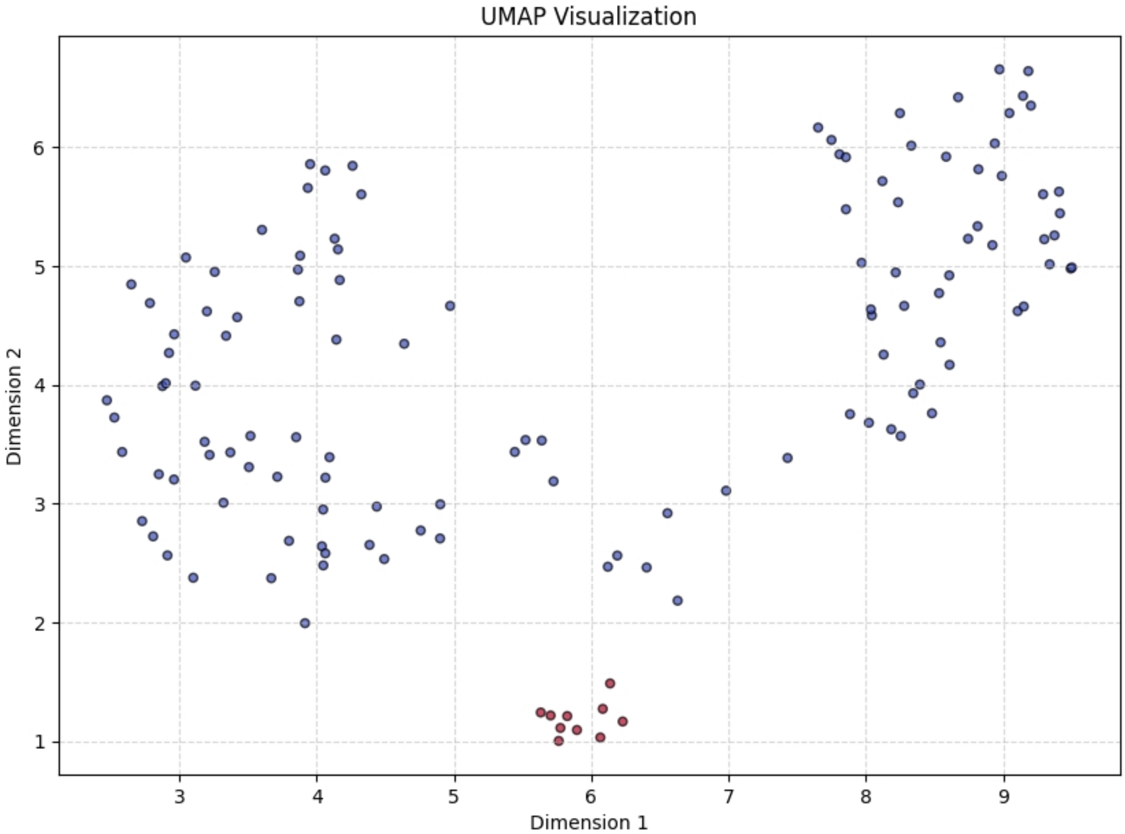

Figure 18: UMAP with $n_{neighbors} = 20$ and $min_{dist} = 0.1$.

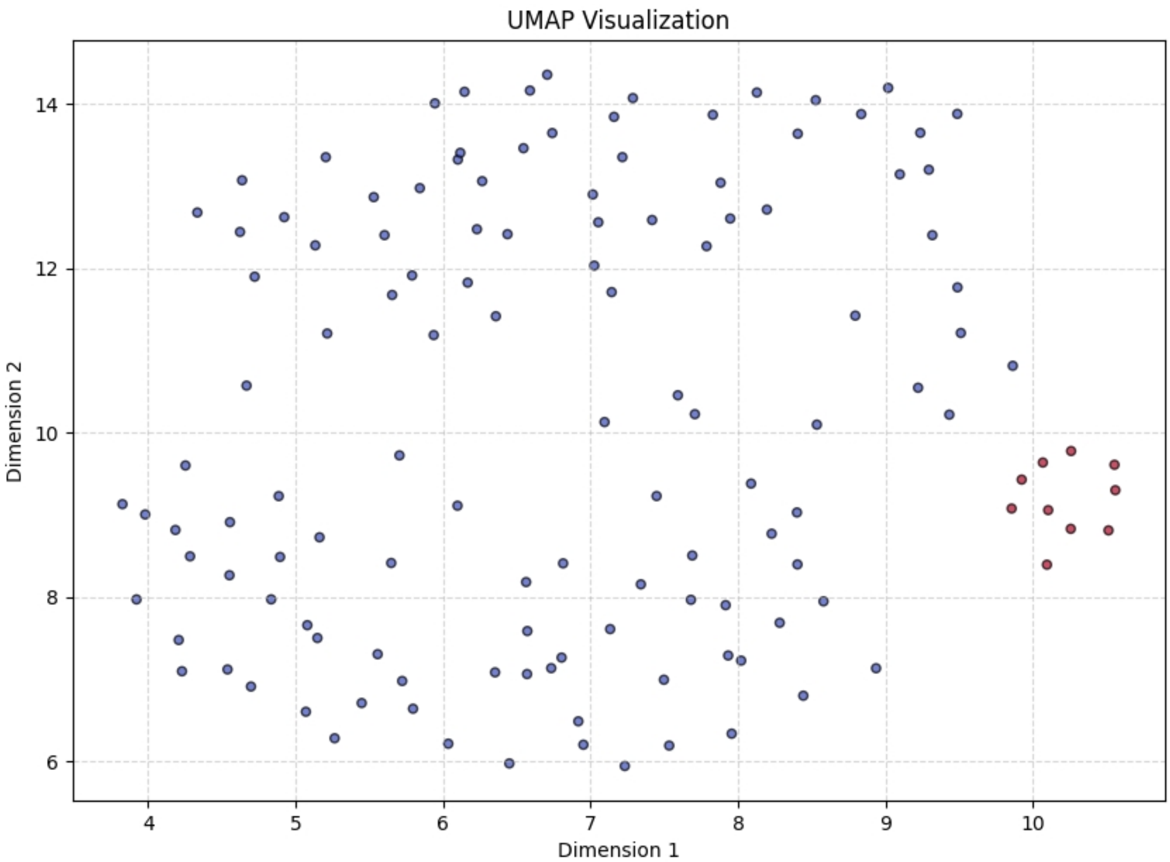

Figure 19: UMAP with $n_{neighbors} = 15$ and $min_{dist} = 0.5$.